

**SafeMedAI: Reducing Harmful Document Exposure in Biomedical RAG
Using a Pre-Reranking Cross-Encoder Filter**

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Chapter 1

THE PROBLEM AND ITS SETTING

Introduction

In the biomedical domain, literature serves as the primary means of disseminating new findings and empirical knowledge, with much of the accumulated research remaining accessible only through published scientific articles (Jin et al., 2024). Consequently, the literature search, the process of retrieving scientific articles to satisfy specific information needs, is critical to all aspects of biomedical research and patient care (Jin et al., 2024). However, the exponential growth of biomedical literature makes identifying relevant information highly challenging. PubMed, the most widely used biomedical literature search engine, currently contains over 36 million articles and continues to expand with the addition of more than 1 million publications annually (Jin et al., 2024).

Historically, platforms like PubMed have functioned as general-purpose search engines that mainly receive short, keyword-based queries from users and return a list of raw articles without further analysis (Jin et al., 2024). Despite significant algorithmic improvements over the past decades, this approach cannot optimally serve specialized information needs, which frequently require alternative query types or enforce highly specific requirements for ranking articles (Jin et al., 2024). To overcome these limitations and better process complex, knowledge-intensive text, natural language processing shifted toward Large Language Models (LLMs) for information retrieval tasks, leveraging their advanced semantic understanding and generation capabilities (Oche et al., 2025; Zhu et al., 2025). Yet, general-purpose LLMs exhibit critical

vulnerabilities when operating solely on their internal parametric memory, most notably the phenomenon of “hallucination,” where the model generates plausible-sounding but factually incorrect or misleading information, alongside an inability to access real-time knowledge or knowledge that has become outdated (Huang et al., 2025; Oche et al., 2025).

To mitigate these parametric limitations, Retrieval-Augmented Generation (RAG) has become the standard paradigm to enhance factual grounding, accuracy, and contextual relevance by explicitly combining LLMs with external information retrieval systems (Lewis et al., 2020; Gao et al., 2024). However, as noted in a comprehensive survey of the domain by Jin et al. (2022), Biomedical Question Answering (BQA) systems remain largely immature and are rarely used in real-life settings. Two difficulties commonly associated with this immaturity are the complex reasoning that biomedical questions demand and the cost of expert-annotated biomedical datasets, which limits how well models can be trained and evaluated in the domain. Consequently, modern RAG pipelines attempt to overcome these retrieval limitations by integrating cross-encoder rerankers to reassess the initially retrieved candidates, ensuring that highly relevant contexts are prioritized (Gao et al., 2024; Glass et al., 2022).

However, critical structural flaws have been identified in standard rerankers. Traditional ranking frameworks optimize exclusively for semantic or lexical similarity between the query and the candidate texts, creating a hidden vulnerability where standard rerankers systematically elevate harmful documents, passages that exhibit strong surface-level alignment with the query but are entirely unsupported or factually incorrect (Liu et al., 2026). As demonstrated by Liu et al. (2026), a counter-intuitive

phenomenon occurs during the second-stage refinement: while standard cross-encoder rerankers successfully improve the recall of ground-truth documents, they simultaneously and significantly increase the proportion of these harmful documents in the top-ranked results. This systematic inflation occurs because traditional rerankers are easily deceived by superficial shortcuts, such as high-frequency entities or overlapping vocabulary, treating dense keyword matching as a proxy for factual truth. Because existing pipelines push relevance and safety through a single, competing ranking objective, the two goals actively trade off against each other. Consequently, standard rerankers fail to suppress these semantically deceptive but logically irrelevant contexts, allowing misleading or counterfactual content to slip directly into the downstream LLM’s limited context window.

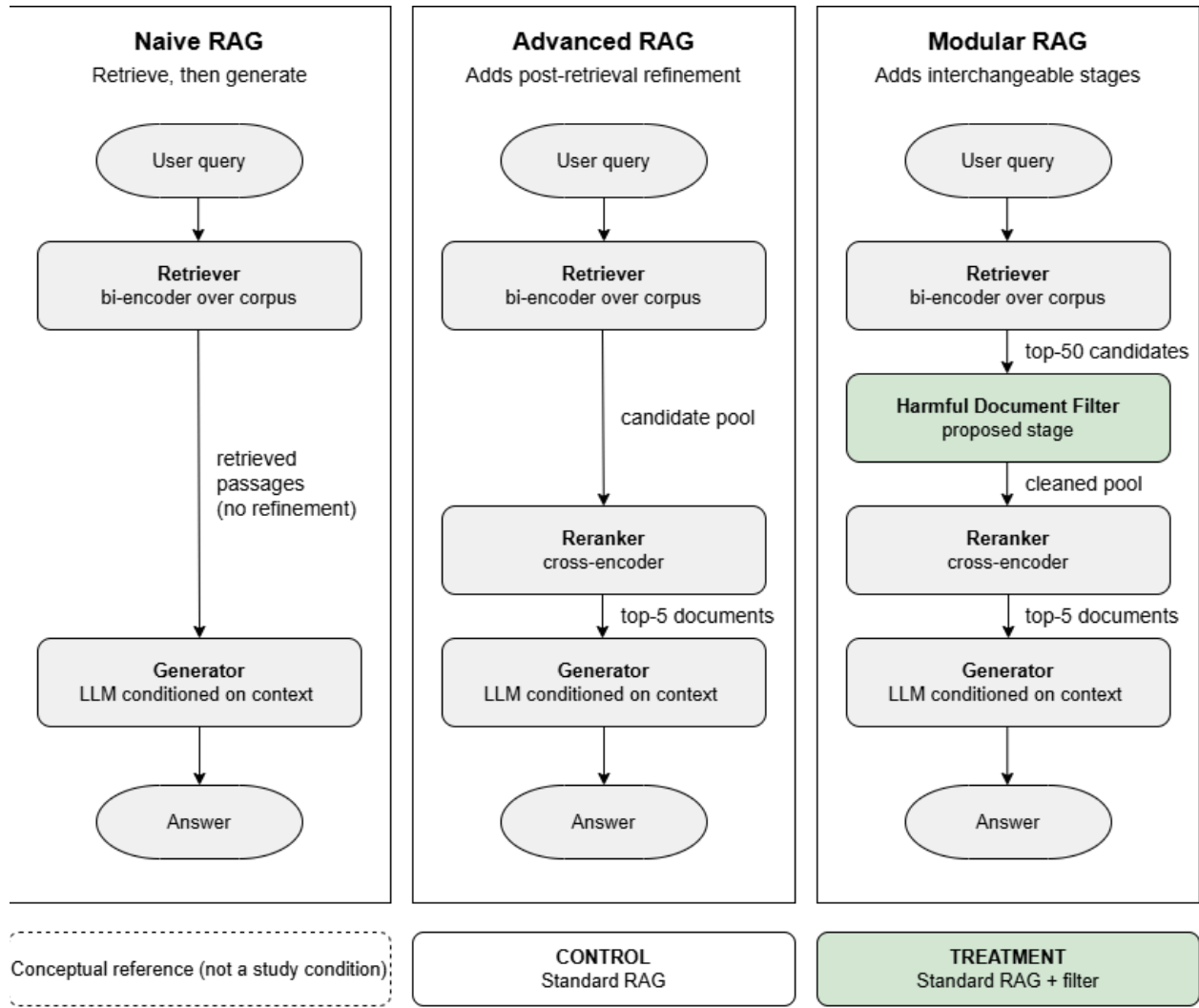
This matters because the composition of the retrieved context shapes what a generator says. Passages that are topically related to a query but do not support the answer degrade generation more than random passages do (Cuconasu et al., 2024), models frequently fail to reject irrelevant or noisy content (Chen et al., 2024), and passages that resemble supporting evidence while contradicting the truth are an adversarial condition that RAG systems must be defended against (Fang et al., 2024). In biomedicine, where clinicians and researchers may act on what a system produces, unsupported output is a safety concern and not merely an inconvenience (He et al., 2025). Existing work has reduced harmful document exposure by redesigning the reranking objective (Liu et al., 2026), but to the best of the researchers’ knowledge, whether a separate stage that removes harmful documents before an unmodified reranker reduces hallucination in the generated answers has not been tested directly.

Therefore, to address this gap, the study proposes a Pre-Reranking Cross-Encoder Filter designed to detect and remove harmful, unsupported documents before the reranking stage, acting as a dedicated safety gate within the retrieval pipeline. The objective of this study is to determine whether adding the filter to a Standard RAG pipeline reduces hallucination in the generated answers, while preserving the genuinely supportive evidence the generator needs. The proposed pipeline is compared with a Standard RAG pipeline that differs from it only in the absence of the filter. Effectiveness is evaluated in three ways: how accurately the filter detects harmful documents (true positive rate, false positive rate, precision, and F1-score); whether the generated answers contain fewer hallucinations (hallucination rate, harmful-document-induced hallucination rate, and answer accuracy); and how the composition of the final context changes (HD@Top-5 and GD@Top-5).

Theoretical Framework

This study is grounded in the theory of Retrieval-Augmented Generation (RAG) and in its progression across three established paradigms, namely Naive RAG, Advanced RAG, and Modular RAG (Gao et al., 2024). These paradigms describe an increasing degree of architectural decomposition, beginning with a minimal retrieve-then-generate sequence and advancing toward pipelines whose individual components are specialized and independently optimizable. Figure 1 situates the proposed framework within this progression by contrasting the three paradigms and identifying the stage at which a dedicated harmful document filter is introduced.

Figure 1. Theoretical Framework



In the Naive RAG paradigm, a retriever selects candidate passages and a generator conditions its output on them directly, without any intermediate refinement (Gao et al., 2024). The Advanced RAG paradigm augments this sequence with post-retrieval refinement, most notably a cross-encoder reranker that reassesses the initial candidates and prioritizes the most relevant contexts (Gao et al., 2024; Glass et al., 2022); this paradigm corresponds to the Standard RAG configuration that serves as the control condition of the present study, in which refinement is concentrated within the ranking stage. The Modular RAG paradigm generalizes the pipeline into a sequence of

interchangeable and individually specialized components, permitting the insertion of new stages without redesigning existing ones (Gao et al., 2024). The proposed framework is positioned within this paradigm and forms the treatment condition, since it introduces a dedicated filtering stage between retrieval and reranking rather than embedding additional behavior inside an existing component.

The framework further rests on the theoretical distinction between the two encoding architectures used in two-stage retrieval. A bi-encoder encodes the query and each candidate independently into a shared vector space, which permits precomputation and rapid nearest-neighbor search but discards the fine-grained interaction between the query and the candidate (Karpukhin et al., 2020; Wang et al., 2020). A cross-encoder, by contrast, jointly encodes the query and the candidate in a single forward pass, modeling their full token-level interaction and thereby achieving higher discriminative accuracy at greater computational cost (Glass et al., 2022). This distinction provides the theoretical justification for the architecture adopted in the study, in which an efficient bi-encoder performs first-stage retrieval over the full corpus while a cross-encoder, applied to a reduced candidate pool, performs the fine-grained judgment required to separate harmful documents from genuinely supportive ones (Glass et al., 2022).

The framework finally rests on the treatment of hallucination in retrieval-augmented systems. Hallucination is the generation of plausible yet nonfactual content, and the literature distinguishes content that contradicts established facts from content that is unfaithful to the supplied context (Huang et al., 2025). Even when a model is grounded in retrieved documents, it may still produce claims that are

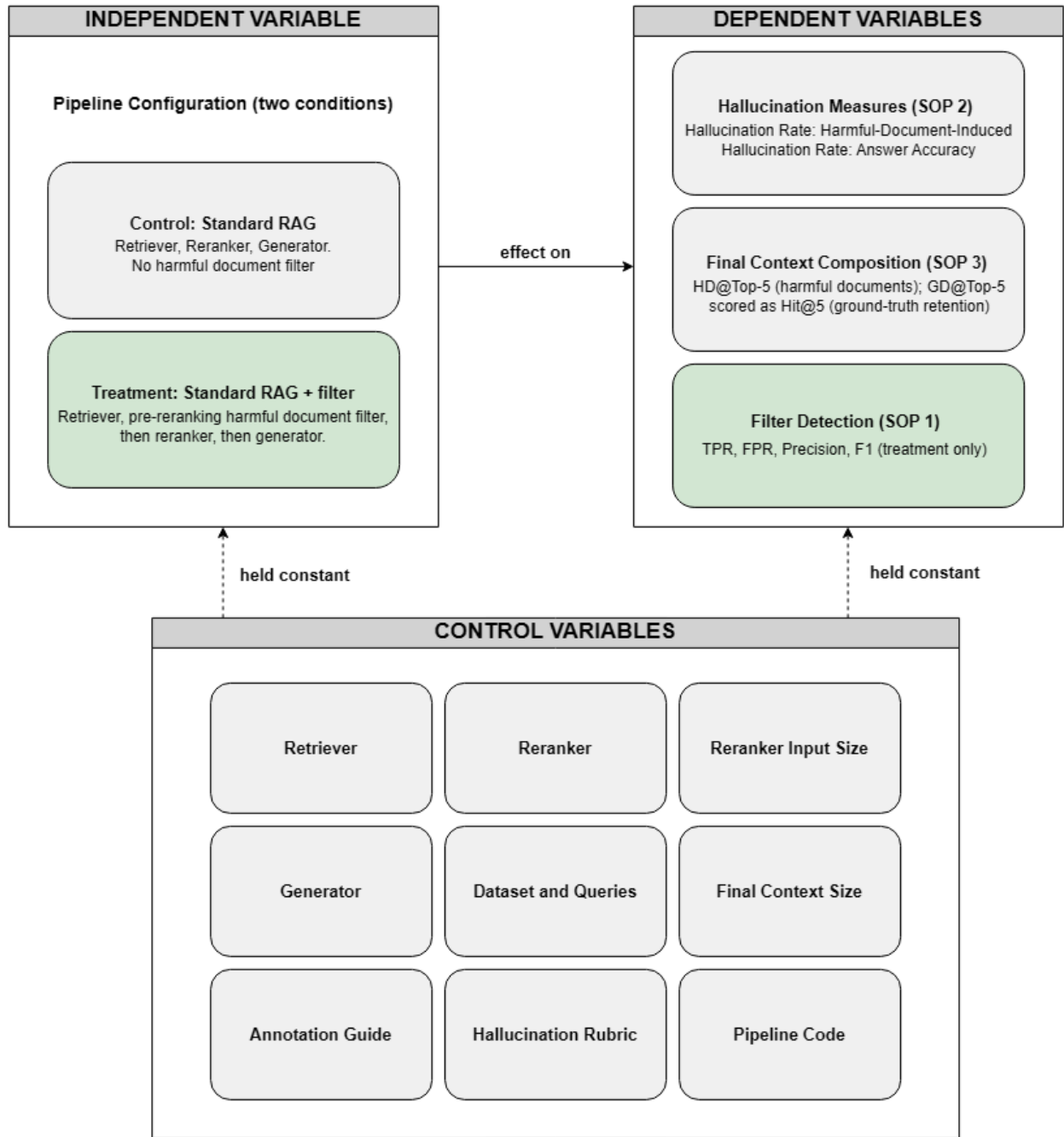
unsupported by, or contradict, the evidence (Niu et al., 2024). Under this view, the retrieved context is a causal input to the answer: if it contains documents that resemble supporting evidence but do not support the correct answer, a generator can reproduce their claims, and the resulting error is a hallucination attributable to the retrieval stage. The study therefore treats harmful documents as the mechanism, the filter as an intervention on that mechanism, and claim-level hallucination in the generated answer as the outcome (Cuconasu et al., 2024; Chen et al., 2024).

Conceptual Framework

The conceptual framework of this study presents the relationship between the pipeline configuration used for biomedical retrieval-augmented generation and hallucination in the generated answers, together with the retrieval-side measures that describe how the filter operates (Figure 2). The independent variable is the pipeline configuration, which has two conditions: Standard RAG (control) and Standard RAG with a dedicated pre-reranking harmful document filter (treatment). The two conditions are compared on hallucination measures and on the composition of the final context, and the detection performance of the filter is reported for the treatment condition.

This framework shows that the study aims to determine whether incorporating a dedicated pre-reranking harmful document filter into a Standard RAG pipeline reduces hallucination in the generated answers compared with the same pipeline without the filter.

Figure 2. Conceptual Framework



Independent Variable

The independent variable of the study is the pipeline configuration used for biomedical retrieval-augmented generation. This refers to how candidate documents are retrieved, filtered, reranked, and passed to the generator component when answering a biomedical query. The two conditions are:

- **Standard RAG with reranker, without a filter (Control).** A bi-encoder retriever passes its candidate pool to a cross-encoder reranker, and the top-ranked documents are passed to the generator. No harmful document filtering is applied. This condition shows the level of hallucination and harmful document exposure that results when relevance reranking is the only refinement applied to the retrieved candidates.
- **Standard RAG with reranker, with the pre-reranking harmful document filter (Treatment).** A fine-tuned cross-encoder classifier is placed between the retriever and the reranker. It screens each retrieved candidate, assigns a probability of being harmful, and removes documents that exceed a set threshold before the reranker processes the remaining pool. The reranker therefore only handles relevance ordering over a pool that has already been cleared of harmful content, rather than being asked to handle both concerns at once. This is the only difference between the two conditions.

Dependent Variables

The dependent variables of the study are the measures used to evaluate the two conditions and the filter itself. They are grouped as follows.

A. Hallucination Measures

- **Hallucination Rate.** The proportion of test queries for which the generated answer contains at least one hallucination, that is, a claim that is unsupported by or contradicts the reference evidence for the query. A lower value indicates that the pipeline is more effective at preventing hallucinated answers.
- **Harmful-Document-Induced Hallucination Rate.** The proportion of test queries whose generated answer contains at least one hallucination that is supported by a harmful document present in the context passed to the generator. This measure links the mechanism of the filter (removing harmful documents) to the outcome (fewer hallucinations).
- **Answer Accuracy.** The proportion of test queries for which the yes or no decision in the generated answer matches the PubMedQA reference label. This measure is supportive, because the labels of the automatically generated subset are heuristic.

B. Final-context Composition Measures. These describe how the context passed to the generator changes and are reported alongside the hallucination measures.

- **Harmful document exposure (HD@Top-5).** The proportion of the five documents passed to the generator that are harmful, meaning they appear relevant to the query based on surface-level similarity but do not support the correct answer and may mislead the generator. A lower value indicates that the pipeline is more effective at protecting the generator from misleading content.

- **Ground-truth retention (GD@Top-5).** Whether genuine evidence that factually supports the correct answer is present among the five documents passed to the generator, scored as a hit when at least one ground-truth document is present (Hit@5). It is evaluated alongside harmful document exposure because a pipeline that removes harmful documents too aggressively may also discard correct evidence. A higher value indicates that the pipeline is preserving the documents the generator needs to construct an answer.

C. Filter detection performance (treatment condition only). With harmful as the positive class, the study reports the True Positive Rate (proportion of harmful documents correctly removed), the False Positive Rate (proportion of non-harmful documents incorrectly removed), Precision, and F1-score.

Control Variables

The control variables of the study are the elements kept constant across both conditions. Holding these variables fixed ensures that any observed differences in the dependent variables can be credited to the pipeline configuration alone and not to other factors that vary between conditions.

- **Retriever model.** The same bi-encoder retriever, all-MiniLM-L6-v2, retrieves the same pool of 50 candidate documents for each query in both conditions. Any difference observed later in the pipeline therefore reflects the effect of filtering, not differences in how documents were initially selected.
- **Reranker model.** The same off-the-shelf cross-encoder reranker, ms-marco-MiniLM-L-6-v2, is used in both conditions, so that performance

differences are due to the presence of the filter and not to a change in the reranker.

- **Reranker input size.** The reranker receives at most 30 candidates in both conditions, so that the size of its input does not differ between the control and the treatment.
- **Generator model, prompt, and decoding parameters.** The same LLM, GPT-4o-mini, the same prompt template, and the same decoding parameters (temperature 0) are used in both conditions, so that differences in the generated answers are attributable to the documents in the context.
- **Dataset and evaluation queries.** Both conditions are evaluated on the same held-out test queries drawn from the pqa_artificial subset of PubMedQA, using the same pooled corpus, so that neither condition is tested on easier or harder queries.
- **Number of documents passed to the generator.** Exactly five documents are passed forward as the final context in both conditions.
- **Annotation guide and labeling criteria.** The same operational definitions of ground-truth, harmful, and mediocre documents are applied during filter training and during evaluation. The criteria used to label a document as harmful during training are the same criteria used to measure HD@Top-5 during testing.
- **Hallucination rubric and judging protocol.** The same rubric, the same judge model, and the same blinded procedure are applied to the answers from both conditions.

- **Pipeline implementation.** Both conditions are run with one implementation of the pipeline, in which the only difference is whether Stage 2 is executed.

Statement of the Problem

This study aims to design, implement, and evaluate a Pre-Reranking Cross-Encoder Filter that mitigates hallucination in biomedical Retrieval-Augmented Generation (RAG) by removing harmful documents from the retrieved candidate pool before reranking.

Specifically, the study seeks to answer the following research questions:

1. What is the performance of the proposed Pre-Reranking Cross-Encoder Filter in detecting harmful documents in terms of:
 - a. True Positive Rate (Recall);
 - b. False Positive Rate;
 - c. Precision; and
 - d. F1-score?
2. Is there a significant difference between Standard RAG with the proposed filter and Standard RAG without the filter against the PubMedQA reference label in terms of:
 - a. Hallucination Rate;
 - b. Harmful-Documents-Induced Hallucination Rate; and
 - c. Answer Accuracy?

3. Is there a significant difference between Standard RAG with the proposed filter and Standard RAG without the filter in the composition of the final context, in terms of:
 - a. Harmful Document Exposure (HD@Top-5); and
 - b. Ground-Truth Retention (GD@Top-5)?

Hypothesis

H₁: The Standard RAG with the proposed pre-reranking filter has a significantly lower Hallucination Rate than Standard RAG without the filter.

H₂: The Standard RAG with the proposed pre-reranking filter has a significantly lower Harmful-Documents-Induced Hallucination Rate and a significantly higher Answer Accuracy than Standard RAG without the filter.

H₃: The Standard RAG with the proposed pre-reranking filter has a significantly lower HD@Top-5 than Standard RAG without the filter.

H₄: There is a significant difference in GD@Top-5 (Hit@5) between Standard RAG with the proposed pre-reranking filter and Standard RAG without the filter.

Scope and Limitation of the Study

This study is scoped to the biomedical question-answering domain and is evaluated on the PubMedQA dataset, specifically its large automatically generated subset (pqa_artificial). The study does not develop or train a language model. It uses established, off-the-shelf components, namely a MiniLM-based bi-encoder retriever, an MS MARCO-trained cross-encoder reranker, and an accessible LLM generator

(GPT-4o-mini), so as to isolate the contribution of the proposed filter rather than the choice of backbone models. The contribution of the study is the treatment pipeline, that is, the filter, together with a validated evaluation framework for measuring hallucination mitigation. The evaluation covers three aspects: the detection performance of the filter, hallucination in the generated answers, and the composition of the final document context. The proposed pipeline is compared against a single control, a Standard RAG pipeline without the filter; other mitigation approaches, including risk-aware reranking, are not benchmarked and are left to future work. The entire pipeline is implemented by the researchers without high-level orchestration frameworks.

Several limitations follow. First, the definitions of harmful, ground-truth, and mediocre documents depend on an annotation procedure that, although validated by human reviewers, relies in part on LLM-assisted labeling and therefore inherits any systematic biases of the labeling model. Second, hallucination is judged against reference evidence by a judge model whose judgments are validated on a human-labeled sample; residual judge error remains possible, and a correct claim that is absent from the reference evidence is counted as unsupported, which is why the generator is instructed to answer only from the provided documents. Third, the yes or no labels of pqa_artificial are assigned by a simple heuristic and are heavily skewed toward “yes” (Jin et al., 2019), so Answer Accuracy is interpreted against the majority-class rate and is treated as supporting evidence only. Fourth, the findings are demonstrated within a single domain, dataset, and generator; generalization to other corpora, open-domain settings, or other generators is not established. Fifth, the study evaluates a fixed set of backbone models and does not exhaustively search the space

of retrievers, rerankers, or generators. Finally, the framework targets harmful documents that arise from topical similarity without factual support; adversarially constructed or maliciously injected content lies outside its scope, and the generated answers are not clinically validated and must not be used for medical decisions.

Significance of the Study

This study contributes to the development of Retrieval-Augmented Generation (RAG) systems by proposing and evaluating a modular pre-reranking harmful-document filtering framework for biomedical RAG pipelines, and by providing a validated way to measure whether such filtering reduces hallucination. The outcomes of this research hold distinct significance for the following stakeholders:

- **Developers of Retrieval-Augmented Generation Systems.** This study provides a practical architectural alternative for mitigating hallucination caused by harmful documents. By introducing a modular filtering stage before reranking, developers may reduce hallucination without redesigning or retraining existing reranking models, allowing harmfulness mitigation to be integrated into existing retrieval architectures.
- **Researchers in Information Retrieval, Retrieval-Augmented Generation, and Biomedical Natural Language Processing.** This study contributes empirical evidence on the relationship between harmful-document exposure and hallucination in biomedical RAG systems. It also evaluates whether adding a dedicated pre-reranking filter reduces hallucination relative to a standard

pipeline, and contributes a validated, claim-level protocol for measuring hallucination attributable to harmful retrieved documents.

- **Developers and Researchers of Biomedical Question-Answering Systems.**

Since biomedical RAG systems rely on retrieved literature as their knowledge source, the findings of this study provide insights into reducing harmful retrieved evidence before answer generation. The proposed framework may support the development of more reliable biomedical research and decision-support applications where the quality of retrieved evidence directly affects generated responses.

- **Future Researchers.** This study establishes a reproducible experimental framework through its human-validated annotation procedure for harmful and mediocre retrieved documents, its human-validated hallucination scoring protocol, and its controlled A/B pipeline design. Future studies may extend this framework by investigating alternative filtering models, reranking architectures, retrieval methods, language models, or application domains requiring trustworthy retrieval.

Definition of Terms

Answer Accuracy. The proportion of test queries for which the yes or no decision in the generated answer matches the PubMedQA reference label.

Claim. A single, self-contained factual statement extracted from a generated answer, used as the unit of hallucination scoring.

Cohen's Kappa (κ). A statistic that measures agreement between two raters on categorical labels beyond what is expected by chance.

Control Condition. The Standard RAG pipeline, which has no harmful document filter.

False Positive Rate (FPR). The proportion of non-harmful (ground-truth or mediocre) candidate documents that the filter incorrectly removes, computed as $FP / (FP + TN)$, where harmful is the positive class.

F1-score. The harmonic mean of precision and true positive rate for the harmful class.

Ground-Truth Document (GD). A retrieved document that factually supports the correct answer to the query.

Hallucination. A claim in a generated answer that is unsupported by, or contradicts, the reference evidence for the query.

Hallucination Rate. The proportion of test queries for which the generated answer contains at least one hallucination.

Harmful Document (HD). A retrieved document that is topically or lexically similar to the query but does not factually support the correct answer, and that may mislead the generator if included in its context.

Harmful-Document-Induced Hallucination. A hallucination whose content is supported by a harmful document present in the context passed to the generator.

Harmful-Document-Induced Hallucination Rate. The proportion of test queries whose generated answer contains at least one harmful-document-induced hallucination.

HD@Top-5 / GD@Top-5. HD@Top-5 is the proportion of the five documents finally passed to the generator that are harmful. GD@Top-5 is ground-truth retention among those five documents, scored as Hit@5.

Hit@5. A per-query score of 1 when at least one ground-truth document appears among the five documents passed to the generator, and 0 otherwise. The mean over queries is the total number of hits divided by the total number of queries.

Judge Model. The language model, different from the generator, that extracts and verifies the claims in generated answers according to the hallucination rubric.

Mediocre Document (MD). A retrieved document that is partially relevant or neutral, neither clearly supporting the correct answer nor actively misleading the generator.

Post-Test Only Control Group Design. An experimental design in which outcomes are measured once, after the treatment, and treated units are compared directly with control units, without a pre-test.

Pre-Reranking Harmful Document Filter. The proposed cross-encoder classifier that screens retrieved candidates and removes high-probability harmful documents before the reranking stage.

Precision. The proportion of documents removed by the filter that are truly harmful, computed as $TP / (TP + FP)$.

Reference Evidence. The evidence against which claims are judged: the PubMedQA context and long answer of the query together with its reference label, and any document in the answer’s own context that is labeled ground-truth.

Reranker. A model, typically a cross-encoder, that reorders an initial pool of retrieved candidates by estimating query–document relevance more accurately than the first-stage retriever.

Retrieval-Augmented Generation (RAG). An architecture in which a language model conditions its output on documents retrieved from an external corpus at inference time, in order to ground generation in evidence and reduce hallucination.

Standard RAG. A RAG pipeline consisting of a retriever, a reranker, and a generator, without a harmful document filter.

True Positive Rate (TPR) / Recall. The proportion of harmful documents that the filter correctly removes, computed as $TP / (TP + FN)$. For the filter, harmful is the positive class: a true positive (TP) is a harmful document removed, a false positive (FP) is a non-harmful document removed, a true negative (TN) is a non-harmful document retained, and a false negative (FN) is a harmful document retained.

Chapter 2

REVIEW OF LITERATURE AND STUDIES

This chapter reviews the body of knowledge that situates the proposed framework within existing research on biomedical retrieval and Retrieval-Augmented Generation. The review is organized thematically rather than as a list of individual works, progressing from the biomedical information-access problem that motivates the study, through the adoption of Large Language Models and the Retrieval-Augmented Generation paradigm that addresses their factuality limitations, to the two-stage retrieval architecture on which the pipeline is built, and then to the harmful document problem that the study directly addresses. It then reviews how hallucination in retrieval-augmented systems can be measured and how the reliability of model-assisted labels and judgments is established, since the study depends on both. The chapter concludes by synthesizing these themes to identify the specific gap that the proposed pre-reranking filter is designed to fill.

Biomedical Literature Retrieval and the Information-Access Problem

Research in the biomedical domain is disseminated primarily through published scientific articles, and the retrieval of those articles to satisfy a specific information need is fundamental to both research and patient care (Jin et al., 2024). The difficulty of this task is driven by scale, since PubMed alone indexes more than thirty-six million articles and continues to grow by over one million publications each year (Jin et al., 2024). A typical query returns hundreds to thousands of articles, yet only a small fraction beyond the first results is ever examined, a pressure that has pushed biomedical search

engines from recency-based ranking toward relevance-based ranking (Jin et al., 2024). The literature therefore frames biomedical information access not as a scarcity of evidence but as the problem of locating the small subset of genuinely supportive evidence within an expanding corpus.

The biomedical question answering literature characterizes this setting as distinctive in both content and difficulty. Surveys of the field describe biomedical question answering as an emerging task that enables applications to access and understand complex biomedical knowledge, while noting that existing systems remain immature and are rarely used in real-life settings (Jin et al., 2022). The PubMedQA benchmark operationalizes this challenge by pairing biomedical research questions with the abstract that addresses them, and it is explicitly constructed so that answering requires reasoning over the quantitative contents of the abstract rather than surface matching (Jin et al., 2019). Its expert-annotated subset contains one thousand instances labeled yes, no, or maybe, whereas its automatically generated subset, pqa_artificial, supplies roughly two hundred eleven thousand question and context pairs whose questions are converted from statement titles and whose yes or no labels are assigned by a simple heuristic; this scale suits training and large-scale evaluation that the smaller expert-annotated subset cannot support, at the cost of noisier labels (Jin et al., 2019).

Domain-specific retrieval models have been developed in direct response to these demands. MedCPT, for example, is trained contrastively on more than two hundred fifty million query and article pairs drawn from PubMed search logs and couples a dense retriever with a re-ranking component, a design that reflects the same

two-stage logic examined later in this review (Jin et al., 2023). It is reported to achieve state-of-the-art zero-shot performance on biomedical retrieval tasks, outperforming general-purpose encoders many times its size (Jin et al., 2023). The recurring lesson across this body of work is that effective biomedical retrieval depends on modeling domain meaning and genuine support rather than lexical overlap, which establishes the motivation, carried through the remainder of this review, for retrieval systems that judge genuine support rather than superficial similarity.

Beyond sheer scale, the biomedical literature is heterogeneous in reliability, since it contains preliminary findings, results that are later contradicted, and conclusions that are subsequently revised, so that two passages addressing the same question may offer opposing support. This heterogeneity is precisely what renders surface similarity an unreliable proxy for factual support in the biomedical domain, and it foreshadows the central difficulty addressed in the present study, namely that a passage may be topically ideal yet evidentially wrong (Jin et al., 2024).

From Lexical Retrieval to LLMs and the Problem of Hallucination

To process complex and knowledge-intensive text beyond the reach of keyword matching, the field has increasingly adopted Large Language Models for information retrieval and question answering, drawing on their capacity for semantic understanding and fluent generation (Zhu et al., 2025; Oche et al., 2025). This shift reframes retrieval as a problem of meaning rather than of surface form, allowing a model to relate a query to a passage even when the two share little vocabulary, and it is the capability that

distinguishes model-based approaches from the lexical systems described in the preceding theme.

This capability is accompanied by a well-documented liability. A comprehensive survey of the phenomenon defines hallucination as the generation of plausible yet nonfactual content and identifies it as a central obstacle to the reliable use of Large Language Models (Huang et al., 2025). When a model relies solely on its internal parametric memory, it is prone to producing confident but incorrect statements and cannot access knowledge that is recent or otherwise absent from that memory (Huang et al., 2025). In the biomedical setting, where an incorrect statement can carry clinical consequences, this liability is especially significant and motivates approaches that ground generation in verifiable external evidence.

The same survey situates retrieval-augmented approaches among the principal strategies for mitigating hallucination, since conditioning a model on retrieved evidence constrains it from relying on unreliable memory (Huang et al., 2025). This positions the following theme, in which the retrieval-augmented paradigm is examined as the field's standard response to the factuality liability of purely parametric models.

The distinction between fluency and factuality is central to this concern. Because Large Language Models are optimized to produce text that is locally coherent and plausible, the surface form of an answer carries little signal as to its correctness, and a confidently phrased statement may be wholly unsupported (Huang et al., 2025). The same property that makes these models attractive for synthesizing biomedical information therefore makes independent grounding indispensable, since the model cannot be relied upon to detect its own errors.

The stakes of this liability are heightened in the biomedical and clinical setting. A survey of retrieval-augmented generation in biomedicine observes that unsupported or fabricated output is not merely an inconvenience in this domain but a safety concern, because clinicians and patients may act on the information a system produces (He et al., 2025). The same literature notes that the knowledge encoded in a model at training time becomes outdated as new evidence is published, leaving purely parametric systems unable to reflect current findings (Huang et al., 2025; He et al., 2025). These domain-specific pressures explain why grounding generation in retrieved, verifiable evidence is treated throughout the biomedical literature as a prerequisite for trustworthy deployment rather than an optional enhancement.

Retrieval-Augmented Generation as a Grounding Paradigm

Retrieval-Augmented Generation was introduced as a means of combining a parametric language model with a non-parametric external memory, so that generation is conditioned on documents retrieved from a corpus at inference time (Lewis et al., 2020). The original formulation demonstrated that grounding generation in retrieved passages improves performance on knowledge-intensive tasks and allows the underlying knowledge to be updated without retraining the model (Lewis et al., 2020). The paradigm has since become a standard response to the factuality limitations of Large Language Models (Gao et al., 2024).

A widely cited survey organizes the development of the paradigm into three progressively more modular forms, namely Naive RAG, Advanced RAG, and Modular RAG (Gao et al., 2024). Naive RAG performs a simple retrieve-then-generate

sequence; Advanced RAG introduces pre-retrieval and post-retrieval refinement, most prominently a reranking stage that reorders the initially retrieved candidates; and Modular RAG decomposes the pipeline into specialized, interchangeable components that can be added or replaced independently (Gao et al., 2024). This progression supplies the conceptual coordinates used in the present study to locate its two conditions, with Standard RAG as the control and the filtered pipeline as a modular stage added to it.

The modular view of the paradigm is significant for the present contribution because it legitimizes the insertion of a new, single-purpose stage. Whereas redesigning a reranker requires retraining and revalidating a component that already performs well at relevance, the modular paradigm permits a dedicated filter to be added between retrieval and reranking without altering the surrounding components (Gao et al., 2024). The proposed framework exploits exactly this property, positioning harmful document filtering as an additional module rather than as a modification of an existing one.

Within biomedicine, the paradigm has been examined directly. A systematic benchmark of retrieval-augmented generation for medicine assembles biomedical corpora, retrievers, and language models, and reports that retrieval augmentation can substantially improve answer quality on medical question answering while also showing that the choice of corpus and retriever materially affects the outcome (Xiong et al., 2024). A broader survey of retrieval-augmented generation in biomedicine catalogs the technologies, datasets, and clinical applications of the paradigm and identifies the reliability of retrieved evidence as a persistent concern for clinical use (He et al., 2025).

Together these works establish both the promise of biomedical RAG and the centrality of evidence quality, which the present study addresses at the retrieval stage.

The evaluation of retrieval-augmented systems has itself become a focus of study, with recent surveys distinguishing the assessment of the retrieval stage from the assessment of the generated answer (He et al., 2025). This distinction matters for the present study because retrieval-side measures alone cannot show whether a change in the retrieved evidence changes what the generator says. The study therefore evaluates both: retrieval-side measures describe how the composition of the evidence supplied to the generator changes, and answer-side measures test whether that change reduces hallucination.

Across these surveys a common limitation is identified. The quality of a retrieval-augmented system is bounded by its first-stage retrieval, which relies on vector or lexical matching that is frequently not precise enough to isolate the most relevant evidence from a large corpus (Gao et al., 2024). Advanced pipelines respond by inserting a reranking stage that reassesses the initial candidates before generation (Gao et al., 2024; Glass et al., 2022), which motivates the closer examination of retrieval and reranking architectures that follows.

Two-Stage Retrieval: Dense Bi-Encoders and Cross-Encoder Reranking

The dominant architecture for first-stage retrieval is the dense bi-encoder, which encodes the query and each candidate independently into a shared vector space and retrieves by nearest-neighbor search. Dense Passage Retrieval established that a learned dual-encoder can outperform strong lexical baselines such as BM25 by a wide

margin on open-domain retrieval, demonstrating the practical viability of dense retrieval (Karpukhin et al., 2020). Because candidate representations can be precomputed, the approach scales to very large corpora; deep self-attention distillation makes the encoders faster still while preserving accuracy, and the compact MiniLM models that result are widely used as efficient retrieval encoders (Wang et al., 2020). Pretrained MiniLM sentence encoders such as all-MiniLM-L6-v2 are distributed through the sentence-transformers framework, which builds on Sentence-BERT (Reimers & Gurevych, 2019). The present study adopts this bi-encoder as its retriever for precisely this balance of speed and accuracy.

The efficiency of independent encoding comes at the cost of interaction, since the query and the candidate are never jointly examined within the model. A cross-encoder removes this limitation by encoding the query and the candidate together in a single pass, modeling their full token-level interaction and achieving markedly higher discriminative accuracy at greater computational cost. Because this cost is prohibitive over an entire corpus, cross-encoders are applied as a second stage to a reduced candidate pool, a division of labor formalized in the retrieve-then-rerank pipeline (Glass et al., 2022). This complementary pairing of efficient first-stage retrieval with accurate second-stage scoring is the architectural foundation on which the present study builds, and it provides the rationale for implementing the proposed harmful document filter as a cross-encoder rather than as a bi-encoder.

The two-stage arrangement is fundamentally a response to a computational constraint. A cross-encoder must process every query and document jointly, so its cost grows with the number of candidates scored, which makes exhaustive cross-encoding

of a large corpus infeasible (Karpukhin et al., 2020). The standard remedy is a funnel, in which an inexpensive first stage reduces the corpus to a manageable candidate pool and successively more discriminating stages operate over progressively smaller sets. The present study follows this principle by retrieving a broad candidate pool with the bi-encoder before applying cross-encoder judgment to the reduced set, which keeps the additional filtering stage computationally tractable (Wang et al., 2020; Glass et al., 2022).

Recent work has extended the reranking stage from supervised cross-encoders to Large Language Models used directly as ranking agents. Instruction-driven listwise reranking with proprietary models has been shown to match or exceed supervised systems on standard benchmarks (Sun et al., 2023), and open, reproducible toolkits now package a range of pointwise, pairwise, and listwise rerankers for use in retrieval-augmented pipelines (Sharifmoghaddam et al., 2025). The present study deliberately fixes a standard, widely benchmarked cross-encoder reranker, trained on the MS MARCO passage-ranking data (Nguyen et al., 2016), rather than adopting a bespoke ranker, so that any observed change in pipeline behavior can be attributed to the proposed filter rather than to the choice of reranker.

Recent work has also refined the ranking stage so that it serves the generator better. RADIO uses a language model to extract the rationales needed to answer a query, and uses them to rerank documents and fine-tune the reranker, aligning the documents the reranker ranks as relevant with those the generator needs (Jia et al., 2025). DynamicRAG treats the reranker as an agent trained with reinforcement learning, using the quality of the language model’s output as feedback to adjust both the

order and the number of selected documents (Sun et al., 2025). CAR uses the change in generator confidence as a signal of document usefulness, promoting or demoting documents in a training-free manner (Song et al., 2026). In the biomedical setting, Vargas and Pedronette (2025) propose a contrastive evidence re-ranking method that fine-tunes embeddings so that factual rationales are separated from misleading explanations, and report initial results on clinical trial reports. These approaches share a structural feature that becomes important in the next theme: they address the mismatch between relevance and usefulness by changing how documents are ranked or how the ranker is trained, within the ranking stage.

A limitation nonetheless runs through this entire body of work. Whether implemented as a supervised cross-encoder or as a language-model ranker, the reranking stage is optimized to estimate query and document relevance, and relevance is operationalized as similarity. The following theme examines what occurs when similarity and factual support diverge.

The placement of a discriminative stage between first-stage retrieval and the generator is itself an established architectural pattern. The retrieve, rerank, and generate framework demonstrated that inserting a neural reranker between a dense retriever and a sequence generator improves performance on knowledge-intensive tasks, and that the reranker can be trained to reorder the retrieved set before it reaches the generator (Glass et al., 2022). This precedent is important for the present study because it establishes that introducing an additional decision stage into the retrieval pipeline is both feasible and effective; the proposed filter extends this pattern by adding

a stage whose objective is the suppression of harmful evidence rather than the maximization of relevance, positioned before the reranker rather than after it.

Noise, Distraction, and Harmful Documents in Retrieval-Augmented Generation

A growing body of work demonstrates that retrieved evidence is not uniformly helpful and that some of it actively degrades generation. A systematic study of how retrieved passages affect a retrieval-augmented system distinguishes relevant, distracting, and random passages, and reports the counter-intuitive finding that distracting passages, which are topically related to the query but do not support the answer, harm the generator more than random passages do (Cuconasu et al., 2024). A complementary benchmark of Large Language Models under retrieval augmentation finds that, as the proportion of noisy passages in the context increases, answer quality declines markedly, and that models frequently fail to reject irrelevant content (Chen et al., 2024).

The most damaging passages are those that combine surface similarity with factual incorrectness. Work on noise-robust training distinguishes relevant, irrelevant, and counterfactual passages and treats the counterfactual case, in which a passage resembles supporting evidence yet contradicts the truth, as an adversarial condition requiring explicit defense (Fang et al., 2024). Related work shows that retrieval-augmented models are not inherently robust to irrelevant context and must be trained or guided to disregard it (Yoran et al., 2024), and that even correctly retrieved evidence can be underused when it is buried within a long context window (Liu et al.,

2024). These findings converge on the conclusion that the composition of the retrieved set, and not merely its average relevance, determines the reliability of the system.

This concern interacts directly with the reranking stage. Because conventional rerankers optimize for similarity, they are liable to promote exactly the distracting and counterfactual passages that are most superficially similar to the query, so that a stage intended to improve evidence quality can simultaneously concentrate harmful evidence at the top of the ranking. The present study refers to such surface-similar but unsupportive passages as harmful documents, a construct that corresponds to the distracting and counterfactual categories described in the robustness literature (Cuconasu et al., 2024; Fang et al., 2024).

Several mitigations have been proposed, and they share a common structural feature. Approaches that unify ranking with generation, training a single model both to rank context and to generate the answer, fold the handling of noise into the ranking or generation objective itself (Yu et al., 2024), and the generator-aware rerankers reviewed in the preceding theme likewise adjust the ranking objective or the training of the ranker to reflect usefulness to the generator (Jia et al., 2025; Sun et al., 2025; Song et al., 2026). The risk-aware reranker of Liu et al. (2026) addresses harmfulness by redesigning the reranking objective to balance relevance against harmfulness within a single component. These solutions demonstrate that harmful document exposure can be reduced, but they do so by concentrating the relevance and safety objectives inside one model, leaving the two objectives entangled.

What this literature does not establish is whether removing harmful documents with a dedicated stage placed before an unmodified reranker reduces hallucination in

the answers that are ultimately generated. The robustness studies show that distracting and counterfactual passages degrade generation (Cuconasu et al., 2024; Chen et al., 2024; Fang et al., 2024), and existing mitigations show that harmful document exposure can be reduced, but the effect of a separate, modular filtering stage on hallucination has not, to the best of the researchers' knowledge, been measured directly. If the promotion of harmful documents results from asking a single similarity-driven component to serve two competing goals, then assigning the safety objective to a dedicated stage before the reranker may reduce harmful document exposure and, in turn, hallucination, without sacrificing the retention of genuine evidence. This unexamined alternative is the gap that the proposed pre-reranking cross-encoder filter is designed to address.

The stakes of harmful document exposure are heightened in the biomedical domain. Because retrieval-augmented systems for medicine are increasingly proposed for research and decision support, a passage that is topically convincing but evidentially wrong can propagate directly into a generated recommendation (He et al., 2025). The hallucination literature similarly notes that misleading retrieved evidence is among the conditions under which grounded models still produce confident errors (Huang et al., 2025). These observations reinforce the rationale for screening harmful documents before they reach the generator rather than relying on the generator to disregard them after the fact.

The composition of the retrieved set also interacts with how a generator uses it. A study of long-context behavior reports that language models attend most strongly to evidence at the beginning and end of the provided context and use evidence placed in the middle far less effectively, so that the position of a passage, and not only its

presence, governs its influence on the answer (Liu et al., 2024). This positional sensitivity strengthens the rationale for evaluating harmful document exposure and ground-truth retention at a fixed top cutoff, because the small set of passages that survive to the top of the ranking is precisely the set the generator is most likely to rely on. Suppressing harmful passages before they reach this limited budget, rather than merely reordering them within it, therefore offers a more direct route to improving the evidence the generator actually consumes.

Measuring Hallucination in Retrieval-Augmented Generation

Measuring hallucination directly is a methodological problem in its own right. Long-form generations typically contain a mixture of supported and unsupported statements, so a single binary judgment of an entire answer is coarse; FActScore addresses this by breaking a generation into atomic facts and computing the percentage of those facts that are supported by a reliable knowledge source (Min et al., 2023). In retrieval-augmented settings, the RAGAs framework defines faithfulness as the extent to which the claims in an answer can be inferred from the retrieved context, and provides reference-free metrics for the retrieval and generation components of a pipeline (Es et al., 2024). Because a generator can be faithful to a harmful document and still be wrong, faithfulness to the retrieved context alone cannot identify hallucination caused by harmful documents, and a reference to trusted evidence is also needed.

RAGTruth, a corpus of naturally generated RAG responses with manual annotation at the case and word levels, characterizes hallucination in this setting as

claims that are unsupported by, or contradict, the retrieved content, and shows that hallucination frequency can be benchmarked and detection methods assessed against human annotation (Niu et al., 2024). The RAG benchmark of Chen et al. (2024) similarly assesses noise robustness and counterfactual robustness by constructing test instances with controlled noisy and counterfactual documents. These works establish two design principles that the present study adopts: hallucination is measured at the level of individual claims, and automated judgments are validated against human labels.

No published benchmark targets the specific question of this study, namely whether removing harmful documents before reranking reduces hallucination on PubMedQA. The study therefore combines a claim-level procedure adapted from the approaches above with a validation protocol against human judgment, described in Chapter 3, and treats the resulting protocol as part of its contribution.

LLM Assisted Annotation and the Reliability of Document Labels and Judgments

Because the present study defines harmful, ground-truth, and mediocre documents through an annotation guide and relies on those labels for both filter training and evaluation, the reliability of large-scale labeling is itself a relevant body of literature. Manual annotation by trained coders or crowd workers is the traditional source of such labels, but it is costly and slow at the scale required to train a classifier (Gilardi et al., 2023).

Recent evidence indicates that Large Language Models can serve as competitive annotators for classification tasks. A widely cited study reports that a general-purpose model performs zero-shot text annotation, including relevance classification, with

accuracy exceeding that of crowd workers and with higher intercoder agreement than both crowd workers and trained annotators, at a small fraction of the cost (Gilardi et al., 2023). This finding supports the use of a language model to generate candidate labels at scale, provided that the resulting labels are validated rather than trusted outright.

The same literature is explicit that model-generated labels must be checked against human judgment (Gilardi et al., 2023). This requirement directly motivates the validation procedure adopted in the present study, in which model-generated labels are compared against independent human labels and accepted only when their agreement, measured by Cohen's Kappa (Cohen, 1960), reaches a predetermined threshold. Agreement at or above 0.80 falls in the range conventionally described as almost perfect (Landis & Koch, 1977). The reviewed literature thus grounds not only the architecture of the proposed system but also the procedure by which its training and evaluation labels are rendered trustworthy.

A further consideration raised by this literature is that the quality of model-assisted annotation depends heavily on the clarity of the instructions supplied to the model (Gilardi et al., 2023). This dependence is the reason the present study fixes a single annotation guide and applies the same operational definitions of harmful, ground-truth, and mediocre documents during both filter training and evaluation, so that the construct learned by the filter is identical to the construct against which it is measured.

The same reliability concern applies to the judgments used to score hallucination. Language-model judges can approximate human judgments closely, but they exhibit systematic biases, including a preference for their own outputs and sensitivity to the

position and length of the answers being compared (Zheng et al., 2023). This is why the present study uses a judge model that differs from the generator, scores answers without knowing which condition produced them and in randomized order, and accepts the judgments only when their agreement with independent human labels reaches a predetermined threshold.

Synthesis of the Study

The reviewed literature traces a single connected line of reasoning. Biomedical knowledge is contained in an enormous and rapidly growing body of articles that keyword-based search cannot navigate adequately (Jin et al., 2024; Jin et al., 2019), which motivated the adoption of Large Language Models for their semantic understanding (Zhu et al., 2025; Oche et al., 2025). That adoption exposed a factuality liability, since models relying on parametric memory hallucinate and cannot reach knowledge absent from that memory (Huang et al., 2025). Retrieval-Augmented Generation answers this liability by grounding generation in retrieved evidence (Lewis et al., 2020; Gao et al., 2024), and within biomedicine it has been shown to improve medical question answering while making the quality of retrieved evidence the decisive factor (Xiong et al., 2024; He et al., 2025). The quality of that evidence is bounded by first-stage retrieval, which is corrected by a reranking stage (Gao et al., 2024; Glass et al., 2022) built from efficient bi-encoders (Karpukhin et al., 2020; Wang et al., 2020) and accurate cross-encoder or language-model rerankers (Sun et al., 2023; Sharifmoghaddam et al., 2025), and refined further by generator-aware ranking (Jia et al., 2025; Sun et al., 2025; Song et al., 2026).

The reviewed works converge on a common strength and a common limitation. Their shared strength is the demonstrated power of relevance modeling to surface semantically appropriate evidence. Their shared limitation, established by the robustness literature, is that relevance modeling optimized for similarity systematically admits and even promotes distracting and counterfactual passages, here termed harmful documents, which degrade generation more than obviously irrelevant content does (Cuconasu et al., 2024; Chen et al., 2024; Fang et al., 2024). Because rerankers optimize for similarity, the very stage meant to improve evidence quality can concentrate harmful evidence at the top of the context.

This convergence also explains why mitigating the problem at the reranking stage alone is difficult. A reranker asked to demote harmful documents must simultaneously continue to promote genuinely relevant ones, and because harmful documents are by construction those that most resemble relevant ones, a single similarity-driven objective cannot cleanly separate the two (Cuconasu et al., 2024; Fang et al., 2024). The present study interprets this difficulty as the practical signature of objective entanglement and as the reason a dedicated, separately optimized stage is worth investigating.

The gap that follows is empirical. The existence of harmful document exposure and the feasibility of mitigating it within a single component are established (Yu et al., 2024; Liu et al., 2026). What the literature does not examine is whether a dedicated stage that removes harmful documents before an unmodified reranker reduces hallucination in the generated answer, and at what cost to the genuine evidence that

survives. The proposed pre-reranking cross-encoder filter is designed to test exactly this.

This synthesis directly informs the methodology that follows. The post-test only control group design, in which the control and treatment pipelines differ by a single component, operationalizes the test of whether the filter itself changes the outcome; the decision to hold the bi-encoder retriever, the standard cross-encoder reranker, and the generator constant follows from the literature that treats these as interchangeable, benchmarked components (Karpukhin et al., 2020; Wang et al., 2020; Sharifmoghaddam et al., 2025), so that any difference is attributable to the filter. The use of a cross-encoder for the filter follows from the evidence that only full query and document interaction can resolve the surface-similar cues that distinguish harmful from supportive passages (Glass et al., 2022; Fang et al., 2024). The evaluation on the pqa_artificial subset of PubMedQA (Jin et al., 2019), and the joint measurement of hallucination, harmful document exposure, and ground-truth retention, follow from the literature that frames the problem as one of composition and downstream effect, in which suppressing harmful evidence must reduce degraded generation without discarding genuine evidence (Cuconasu et al., 2024; Chen et al., 2024). The claim-level measurement of hallucination follows from the approaches reviewed in the measurement theme (Min et al., 2023; Niu et al., 2024).

Viewed together, the reviewed themes also clarify why the study measures success along three axes rather than one. The robustness literature establishes that a retrieval set is characterized by the amount of harmful evidence it contains and the amount of genuine evidence it preserves, that intervening on one can unintentionally

affect the other (Cuconasu et al., 2024; Fang et al., 2024), and that the consequence of harmful evidence is degraded generation (Cuconasu et al., 2024; Chen et al., 2024). A filter that removed harmful passages but also discarded supportive ones would not represent progress, and a filter that changed retrieval composition without reducing hallucination would not achieve its purpose. The study therefore measures hallucination directly, and reports harmful document exposure and ground-truth retention alongside it, together with the detection performance of the filter itself.

Finally, because filter training, evaluation, and hallucination scoring depend on labels and judgments produced with language models, their reliability is secured through human-validated procedures, consistent with evidence that language-model labeling can match human annotation when checked against it (Gilardi et al., 2023) and that the biases of language-model judges must be controlled (Zheng et al., 2023). This completes the bridge from the reviewed literature to the research design presented in Chapter 3.

CHAPTER 3

METHODOLOGY

This chapter describes the methods used to design, implement, and evaluate the Pre-Reranking Cross-Encoder Filter. It presents the research design, the sources and sampling of data, the architecture of the pipeline together with the software tools and models used to build it, the formal problem definition, the research instruments, the procedure for generating and gathering data, the ethical considerations, and the plan for data analysis and statistical treatment. The chapter is organized so that each research question stated in Chapter 1 maps to a measure, a procedure, and an analysis.

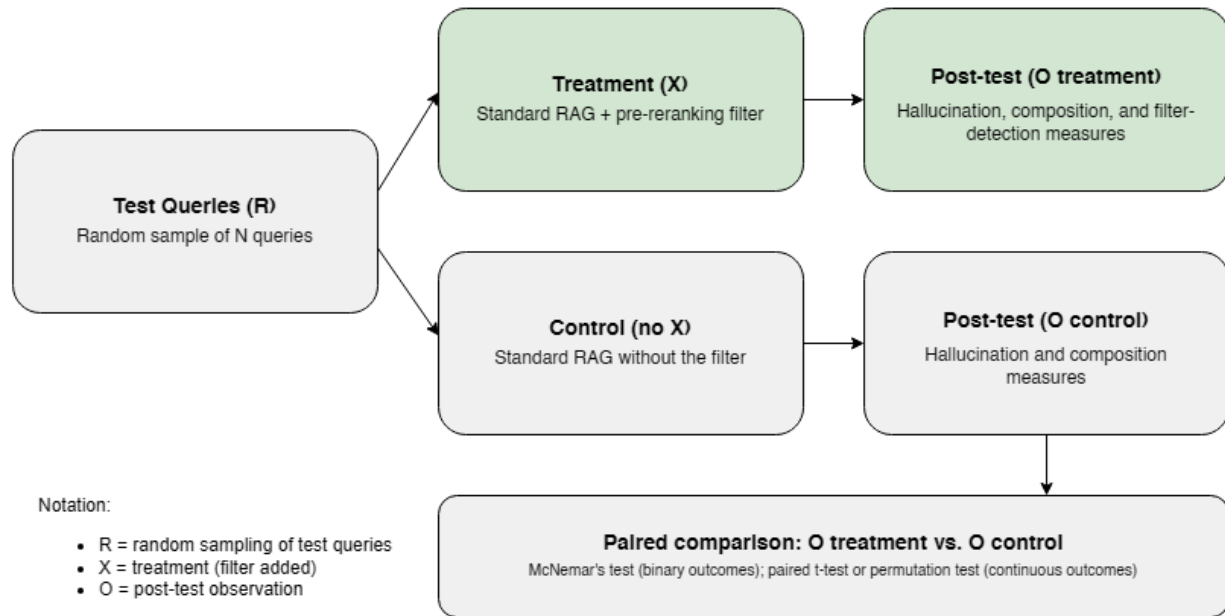
Research Design

The study employs a post-test only control group experimental design. The treatment is the addition of the pre-reranking harmful document filter to a Standard RAG pipeline, and the control is the same pipeline without the filter. Outcomes are measured once, after each pipeline has processed a query, and the two conditions are compared directly. No pre-test is administered, because the outcomes of interest, such as hallucination in a generated answer, exist only after the pipeline has run.

The unit of analysis is the query. Every test query is processed by both pipelines, so the design is a matched-pairs version of the post-test only control group design: each query serves as its own matched control, the two outcomes for a query form a pair, and the comparison uses paired statistical tests. Test queries are drawn at random from the held-out portion of the dataset. Because every query receives both conditions, no

random assignment of queries to conditions is needed. Figure 3 presents the design in the notation of Campbell and Stanley.

Figure 3. *Post-test only control group experimental design*



The design is experimental in a computational sense. The researchers manipulate the independent variable, which is whether Stage 2 (the filter) is executed, hold every other component constant as specified under Control Variables in Chapter 1, and apply both conditions to the same test queries through a single implementation of the pipeline. The filter is developed separately, on training and validation queries, and never sees a test query. The main threats to validity and the corresponding safeguards are as follows.

- **Selection and query difficulty.** Both conditions are run on the same test queries, and the analysis is paired by query.

- **Instrumentation.** The same annotation guide, rubric, and judge model are used for both conditions; answers are coded, stripped of condition identifiers, and scored in randomized order.
- **Data leakage.** Training, validation, and test queries are disjoint at the query level, and the threshold, the prompt, and the analysis plan are fixed before the test queries are run.
- **Generator nondeterminism.** The generator runs at temperature 0, and a regeneration check on a subset of queries measures residual variation.
- **Construct validity.** Hallucination is measured at the claim level and the judge model is validated against independent human labels (see Data Analysis).
- **External validity.** Findings are limited to the dataset, domain, and generator used; this is acknowledged in the Scope and Limitation of the Study.

Sources of Data

The study uses the pqa_artificial subset of PubMedQA (Jin et al., 2019). Each instance consists of a question that is derived from an article title, a context that is the article’s abstract without its conclusion, a long answer that is the conclusion of the abstract, and a yes or no label that is assigned by a simple heuristic. The subset contains roughly 211,000 instances, of which 92.8% carry the label yes and 7.2% the label no (Jin et al., 2019). The contexts of all instances are pooled into a single corpus that serves as the knowledge base for retrieval. The question of an instance is used as a query, and the long answer, the label, and the query’s own context together form the

reference evidence for that query. The dataset is publicly available under the MIT license and contains no patient data.

Sampling of Data

Queries are partitioned at the query level into three non-overlapping sets by simple random sampling with a fixed random seed (seed = 42): training queries ($N_{\text{train}} = 1,000$), validation queries ($N_{\text{val}} = 250$), and test queries ($N_{\text{test}} = 500$). Because the corpus is pooled from all instances, the partition determines how queries are used and not which documents can be retrieved. The training queries are used to fine-tune the filter. The validation queries are used to select the filter checkpoint, to tune the threshold, and to calibrate the annotation guide and the hallucination rubric. The test queries are used only to run both pipelines and to evaluate them.

The number of test queries ($N_{\text{test}} = 500$) is determined by a power analysis for McNemar's test at a two-sided significance level of $\alpha = 0.05$ and a power of $1 - \beta = 0.80$, assuming a smallest effect of interest of 10 percentage points (0.10) in Hallucination Rate and an anticipated proportion of discordant queries of 0.25. This setup yields a theoretical minimum sample size of $n = 194$ pairs; therefore, utilizing 500 queries provides conservative statistical power exceeding 0.95 and absorbs variance. The proportion of test queries whose reference label is "no" is reported (7.2%).

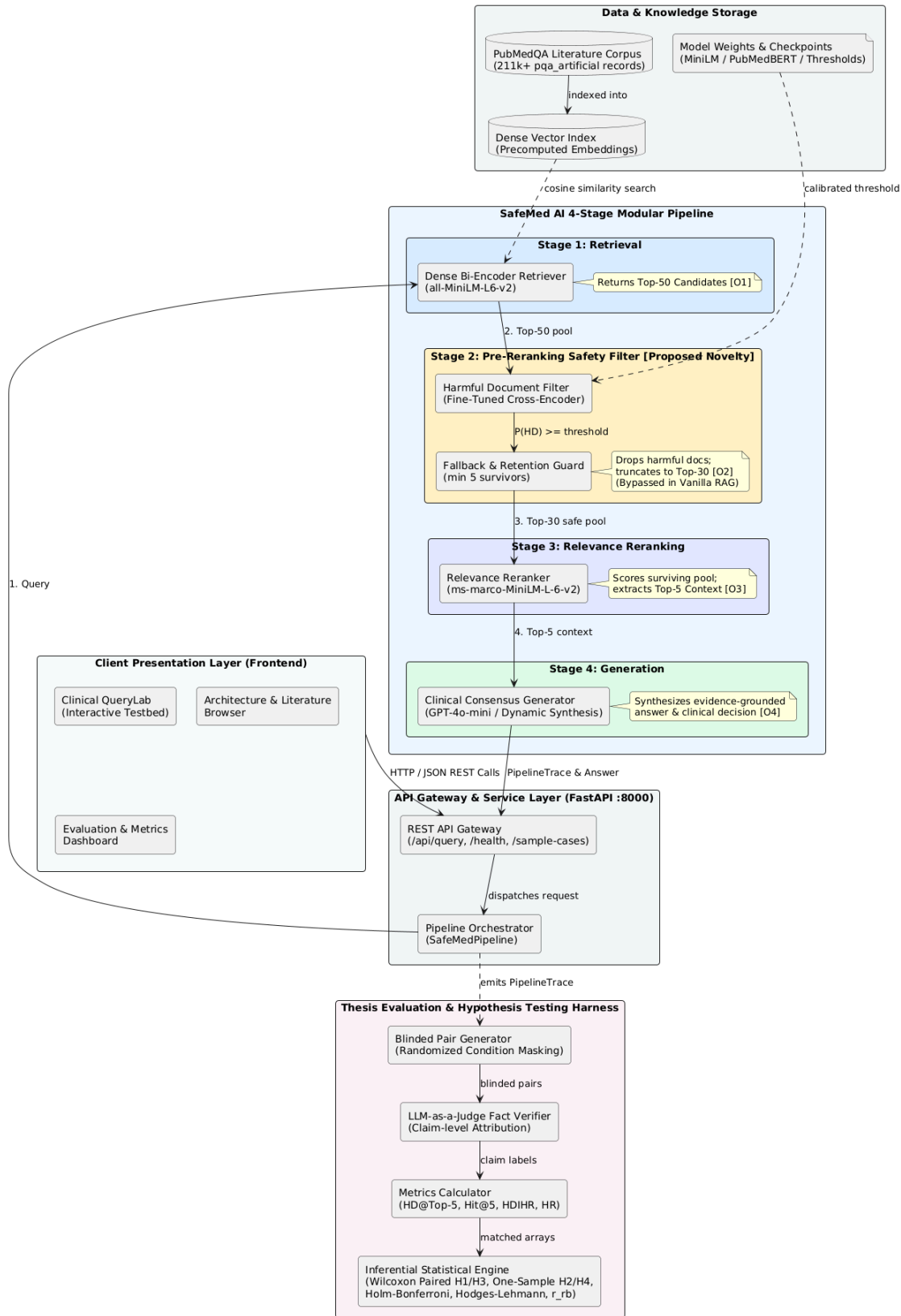
For annotation, $n_{\text{annot}} = 1,500$ candidate pairs drawn from the training and validation queries are labeled, stratified by retriever rank so that both highly ranked and lowly ranked candidates are included; an independent subset of $n_{\text{human}} = 300$ pairs is labeled by human reviewers to measure agreement. All 50 candidates of every test

query are labeled by the validated labeling procedure, because the evaluation of the filter (SOP1) and of the final-context composition (SOP3) requires a label for every test candidate. For the validation of hallucination scoring, `nval_answers` = 100 answers per condition (200 answers in total) are scored independently by human reviewers.

System Architecture

The system consists of the five-stage pipeline shown in Figure 4. Stages 1 to 4 form the retrieval-augmented pipeline that produces answers. Stage 5 is the evaluation layer used by the researchers to measure outcomes and is not part of the pipeline itself. The pipeline is implemented in strict one-to-one adherence to the system class diagram (SafeMedAIPipeline, Retriever, HarmfulDocumentFilter, Reranker, Generator, and JudgeEvaluator). Every query executes Stages 1 to 4 in sequence, and the trace artifacts O1 to O4 are recorded at each stage boundary. The pipeline is implemented from scratch by the researchers in Python, without high-level orchestration frameworks such as LangChain or LlamaIndex; low-level libraries for loading models and for numerical computation are used as software tools (Table 1).

Figure 4. System Architecture Diagram



Software Tools and Models

Table 1 lists the software tools and models used in the study. They are the means by which the pipeline is built and run and are distinct from the research instruments, which are described in a later section.

Table 1. Software Tools and Models

Category	Item	Role in the study
Programming language	Python 3.10	Implementation of all pipeline logic, run scripts, and analysis
Model libraries	PyTorch, Hugging Face Transformers, and sentence-transformers	Loading pretrained models and fine-tuning the filter
Vector search	FAISS (faiss-cpu) with exact cosine similarity fallback via NumPy	First-stage retrieval over corpus embeddings
Retriever model	all-MiniLM-L6-v2 (Reimers & Gurevych, 2019; Wang et al., 2020)	Stage 1: bi-encoder embedding of queries and corpus
Filter model	Cross-encoder (cross-encoder/ms-marco-MiniLM-L-6-v2) with a three-class head, fine-tuned by the researchers	Stage 2: harmful probability for each (query, document) pair
Reranker model	ms-marco-MiniLM-L-6-v2 (trained on MS MARCO; Nguyen et al., 2016), not fine-tuned	Stage 3: relevance reranking
Generator model	GPT-4o-mini gpt-4o-mini-2024-07-18; accessed March 2026	Stage 4: answer generation; also the candidate labeler for document labels

Judge model	GPT-4o (gpt-4o-2024-08-06)	Stage 5: claim extraction, claim verification, and attribution
Statistical software	SciPy (scipy.stats) and statsmodels (supplemented by NumPy and scikit-learn)	Wilcoxon signed-rank tests with Pratt's zero-method and agreement statistics

Stage 1: Retriever

The retriever embeds the query and every context in the pooled corpus using the all-MiniLM-L6-v2 bi-encoder. Corpus embeddings are computed once and stored. For each query, the retriever computes the cosine similarity between the query embedding and all corpus embeddings and returns the 50 highest-scoring documents in descending order of similarity. The same 50 candidates, with their similarity scores, are passed to both conditions.

Stage 2: Pre-Reranking Harmful Document Filter (Treatment Only)

The filter is a cross-encoder classifier that takes a (query, document) pair as one input sequence and outputs probabilities over three classes: ground-truth (GD), mediocre (MD), and harmful (HD). The harmful probability, $P(\text{harmful} \mid q, d)$, is the probability assigned to the HD class. The filter is fine-tuned on labeled pairs from the training queries with a class-weighted cross-entropy loss to counter the imbalance among the classes, and the fine-tuning is repeated over three random seeds.

At test time, all 50 candidates of a query are scored. Candidates whose harmful probability exceeds the threshold τ are removed. The remaining candidates are ordered by retriever similarity and the top 30 are passed to the reranker. If fewer than five

candidates remain after removal, the removed candidates with the lowest harmful probability are reinstated until five are available, and the number of reinstatements is reported. The threshold τ is selected on the validation set to maximize the F1-score of the harmful class and is fixed before any test query is run.

To prevent downstream information starvation, the filter employs a Safety Fallback Guard. Let S denote the set of surviving, unflagged candidates. If $|S| \geq 5$, all flagged documents are permanently discarded. If $|S| < 5$, the lowest-risk flagged candidates are iteratively reinstated into S until exactly five documents are secured, with all reinstatements logged in the audit trace. Finally, the surviving candidates are ordered by retriever similarity and the top 30 are passed to the reranker.

Stage 3: Reranker

The reranker is an off-the-shelf cross-encoder that is not fine-tuned. It receives at most 30 candidates in both conditions: in the control condition, the top 30 of the 50 candidates by retriever similarity, and in the treatment condition, the top 30 of the candidates that survive the filter. Capping the input in both conditions ensures that the size of the pool the reranker sees does not differ between the conditions. The reranker scores each (query, document) pair, sorts the pool in descending order of score, and keeps the top five documents.

Stage 4: Generator

The generator, GPT-4o-mini, receives the query and the five documents in reranker order, numbered 1 to 5, without any label or indicator of the condition. An

identical prompt is used in both conditions. It instructs the model to answer using only the provided documents, to state when the documents are insufficient, to begin with “Decision: yes”, “Decision: no”, or “Decision: maybe”, and then to give a brief explanation of no more than 150 words. The temperature is 0. The prompt template is fixed before the test queries are run.

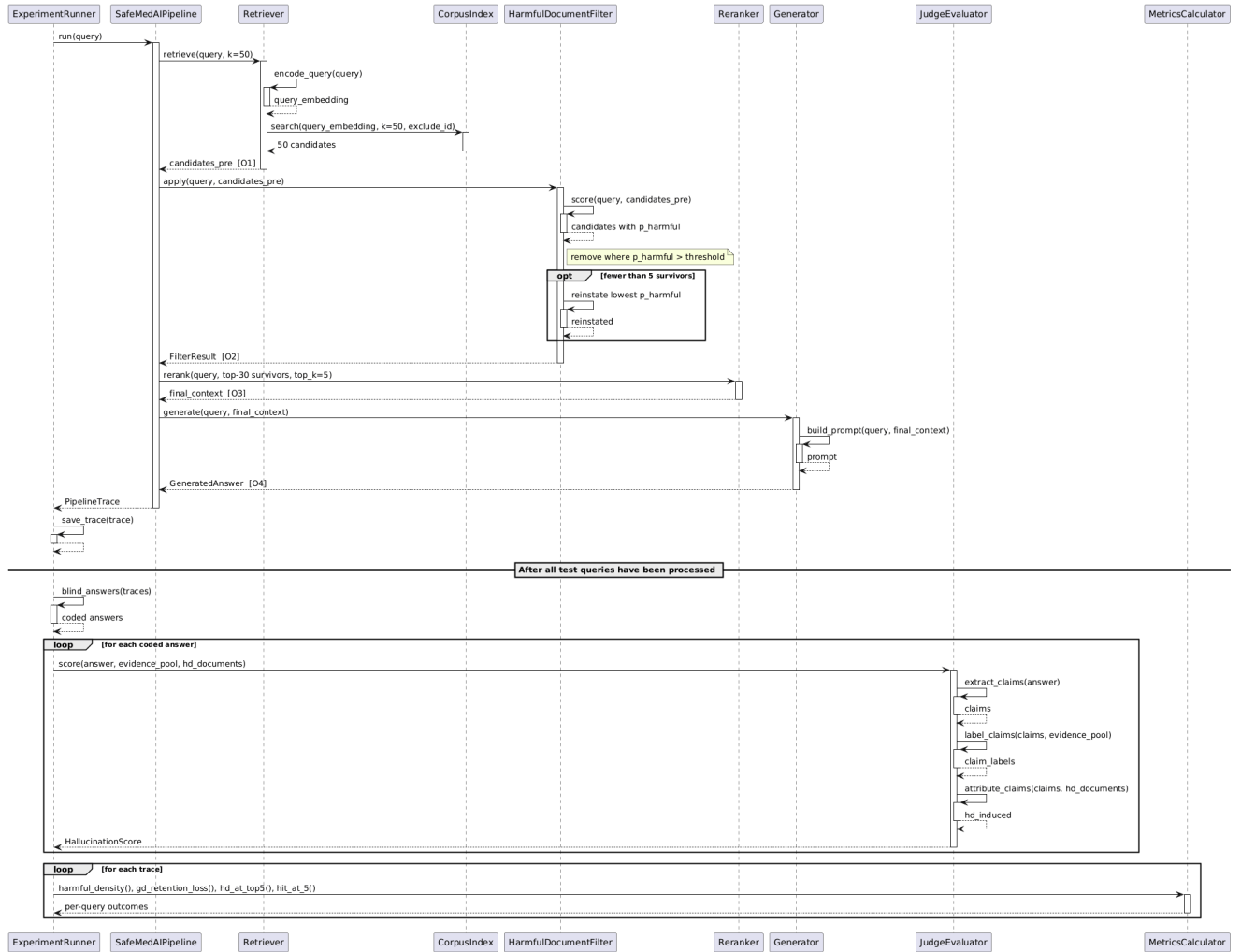
Stage 5: Evaluation

The evaluation layer takes the outputs of the pipelines and computes the measures defined in Table 3. It compares the filter decisions with the document labels (SOP1), applies the blinded hallucination scoring protocol to the generated answers (SOP2), compares the five documents passed to the generator with the document labels (SOP3), and runs the paired statistical analysis described under Data Analysis.

Operational Flow of the Pipeline

Figure 5 is a UML sequence diagram of one experimental run for one query. The Experiment Runner retrieves the candidates once, runs the control and treatment branches in parallel from the same 50 candidates, and then sends the coded answers to the judge model for blinded scoring and attribution.

Figure 5. Sequence Diagram



Figures 6 and 7 are UML activity diagrams of the two phases of the study. Figure 6 shows the development phase, in which the corpus is prepared, the queries are partitioned, documents are annotated until the agreement threshold is met, and the filter is trained and its threshold frozen. Figure 7 shows the experimental phase, in which every test query is run through both conditions, the answers are scored by the judge model and validated against human labels, and the analyst computes the outcomes and runs the pre-specified tests.

Figure 6. Activity Diagram of the Development Phase

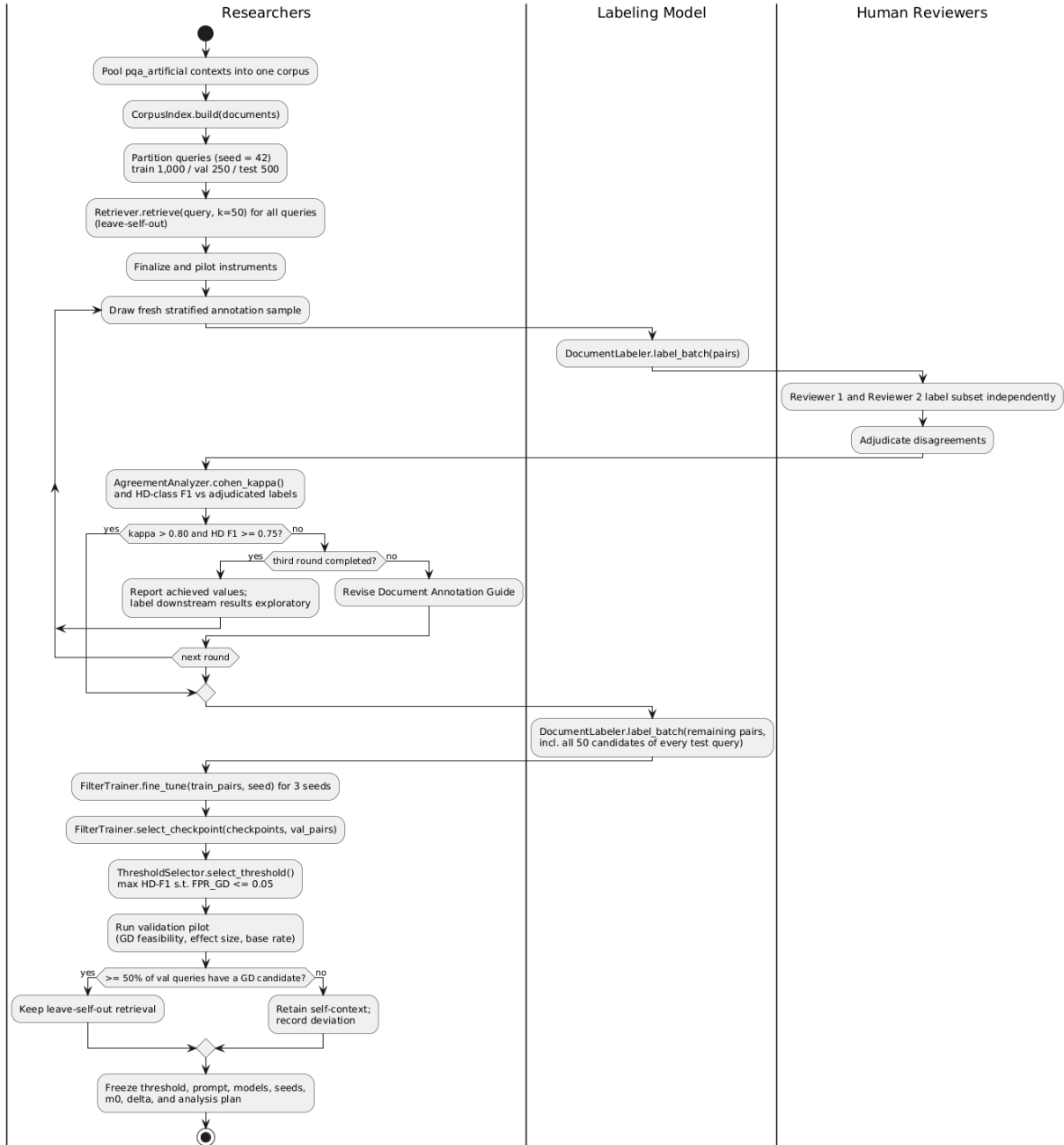


Figure 7. Activity Diagram of the Experimental Phase

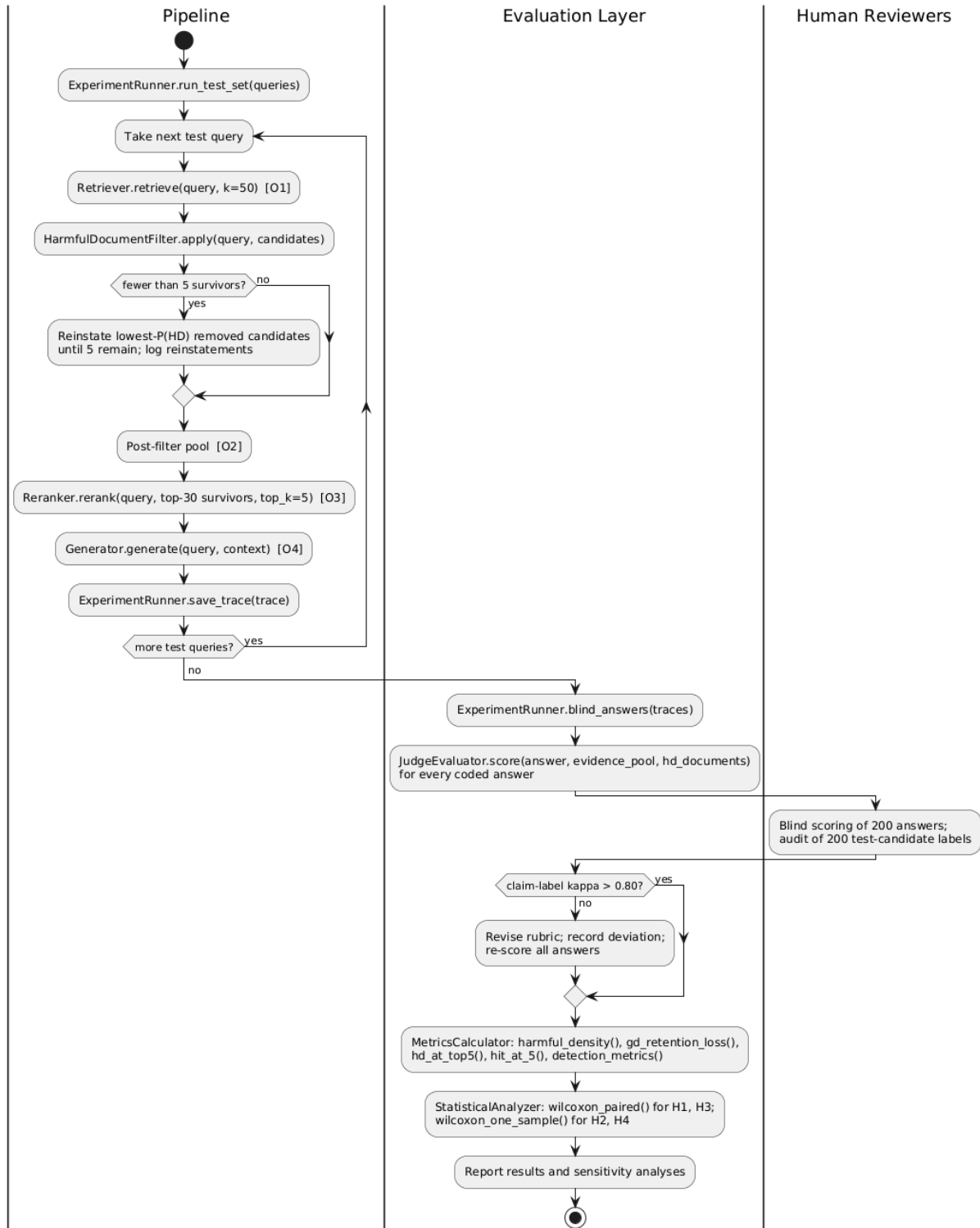
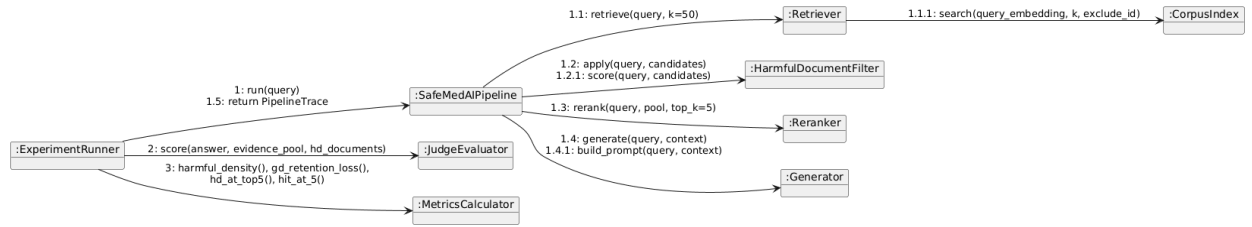


Figure 8. Class Diagram



Figure 9. Collaboration Diagram



Formal Problem Definition

Let Q be the set of test queries and D the pooled corpus. For a query q in Q , the retriever returns the ordered candidate set $C_q = \{d_1, \dots, d_{50}\}$, ordered by the similarity score $s(q, d)$. Each candidate has an annotated label $y(q, d)$ in $\{GD, MD, HD\}$. The filter is a function $f(q, d) = P(\text{harmful} \mid q, d)$, and τ is its threshold.

- **Control pool:** $P_q(C) =$ the 30 candidates of C_q with the highest $s(q, d)$
- **Treatment pool:** $R_q = \{d \text{ in } C_q : f(q, d) \leq \tau\}$; $P_q(T) =$ the 30 candidates of R_q with the highest $s(q, d)$

In each condition, the reranker g orders the pool and the final context K_q consists of the five highest-scoring documents. The generator produces an answer $a_q = \text{LLM}(q, K_q)$, so

that each query yields two answers, $a_q(C)$ and $a_q(T)$. The outcomes are then defined as follows.

- $H(a) = 1$ if a contains at least one unsupported or contradicted claim, otherwise 0
- $H_{HD}(a) = 1$ if $H(a) = 1$ and a hallucinated claim is supported by a harmful document in K_q , otherwise 0
- $A(a) = 1$ if the decision in a equals the reference label, otherwise 0
- $HD@Top-5(q) = (\text{number of documents } d \text{ in } K_q \text{ with } y(q, d) = HD) / 5$
- $Hit@5(q) = 1$ if at least one document d in K_q has $y(q, d) = GD$, otherwise 0
- The objective is to determine, over the test queries, whether the probability that $H(a_q(T)) = 1$ is lower than the probability that $H(a_q(C)) = 1$ (H1), and to describe how H_{HD} , A , $HD@Top-5$, and $Hit@5$ differ between the conditions (H2 to H4) and how accurately f identifies the candidates labeled HD (SOP1).

Research Instrument

The research instruments of this study are the rubrics, guides, and sheets used to define, record, and score the data. Table 2 summarizes the instruments, and their outlines are provided in Appendix B.

Table 2. Research Instrument

Instrument	Purpose and content	Used by and when
1. Document Annotation Guide (codebook)	Defines ground-truth (GD), harmful (HD), and mediocre (MD) documents, with decision rules, boundary cases, and worked examples	LLM labeler and human reviewers; document labeling for training and evaluation

2. Document Annotation Sheet	Records, for each (query, document) pair, the reference answer, the label from each rater, a rationale, and the adjudicated label	Human reviewers; agreement measurement
3. Hallucination Scoring Rubric	Defines a claim, the labels Supported, Unsupported, and Contradicted, the answer-level hallucination flag, the treatment of the decision line, and the attribution rule for harmful-document-induced hallucination	Judge model and human reviewers; scoring of generated answers
4. Hallucination Scoring Sheet	Records, for each coded answer, the extracted decision, each claim with its label, the answer-level flag, the attribution result, and the rater	Judge model (outputs) and human reviewers; scoring and validation

The annotation guide and the hallucination rubric are piloted on a small set of validation pairs and answers before use, and their wording is refined until the reviewers report no unresolved ambiguity. Their reliability is then quantified through inter-rater agreement, as described under Data Analysis.

Data Generation and Gathering Procedure

Data is generated by running the pipelines on the test queries and are gathered through the instruments above. The procedure is carried out in the following steps.

1. **Prepare the corpus and the query partitions.** Pool the contexts of pqa_artificial into one corpus, embed and index it once, and partition the queries into training, validation, and test sets with a fixed random seed.
2. **Retrieve candidates.** For every query in the three sets, retrieve the top 50 candidates with their similarity scores and store them.
3. **Finalize and pilot the instruments.** Complete the annotation guide, the annotation sheet, the hallucination rubric, the scoring sheet, and the protocol sheet, and pilot them on a small set of validation pairs and answers.
4. **Annotate documents.** The labeling model labels a stratified sample of pairs using the annotation guide. Independent human reviewers label a subset, and Cohen's Kappa between the labeling model and the reviewers is computed. If the Kappa is below 0.80, the guide is revised and the procedure is repeated on a new sample. When the threshold is met, the remaining pairs, including all 50 candidates of every test query, are labeled.
5. **Train and tune the filter.** Fine-tune the filter on the training pairs over three random seeds, select the checkpoint and the threshold τ on the validation set, and freeze them.
6. **Fix the run parameters.** The prompt template, model versions, decoding parameters, seeds, and the analysis plan are finalized before any test query is run.
7. **Run both pipelines on the test queries.** For each query, retrieve once and pass the same 50 candidates to both branches. Log the filter probabilities, the pools, the final five documents, the prompt, the answer, and the timestamp.

- 8. Blind and randomize the answers.** Assign each answer a random code, remove condition identifiers, and randomize the order of presentation to the judge.
- 9. Score hallucination.** The judge model extracts the claims of each answer, labels each claim against the reference evidence, and sets the answer-level flag. In a separate second pass, it attributes each hallucinated claim to the harmful documents in that answer's context.
- 10. Validate the scoring.** Human reviewers, blind to the condition, score a random sample of answers, and the agreement between the judge model and the reviewers is computed. If agreement is below the threshold, the rubric is revised, the revision is recorded as a deviation, and the answers of both conditions are re-judged.
- 11. Run the reliability checks.** Re-score a subset with the judge model, regenerate answers for a subset, and compare the filter across its three seeds.
- 12. Analyze.** Compute the outcomes and apply the pre-specified tests described under Data Analysis.

Ethical Considerations

The study uses a publicly available dataset of biomedical research abstracts and involves no patients, clinical records, or personal data. The texts sent to the language model services are public research abstracts and questions derived from article titles. Human reviewers take part in labeling and scoring voluntarily, are informed of the task, and provide no personal information; their labels are recorded under anonymous

identifiers. The study will undergo the institutional ethics review process (UREC) before data collection. The generated answers are outputs of a research prototype, are not validated clinically, and must not be used for medical decisions; this is stated in the limitations of the study. To protect the integrity of the findings, the analysis plan, threshold, and prompt are fixed before the test queries are run, and all results, including those that do not support the hypotheses, are reported.

Data Analysis

Quality of Annotation and Judgment

The quality of the document labels and of the hallucination judgments is established before the outcomes are analyzed. Agreement between the labeling model and the human reviewers on the labels GD, HD, and MD is measured with Cohen's Kappa (Cohen, 1960), and the confusion matrix is reported. Agreement between the judge model and the human reviewers is measured at two levels: the label of each claim (Supported, Unsupported, Contradicted) and the answer-level hallucination flag. For both, a Kappa of at least 0.80 is required, a level conventionally described as almost perfect agreement (Landis & Koch, 1977); when the threshold is not met, the guide or the rubric is revised as described in the procedure. For the answer-level flag, the F1-score of the judge against the human labels is also reported, with hallucinated as the positive class.

Outcome Measures and Their Scoring

Table 3 gives the operational definition and the level of measurement of each outcome. All outcomes are computed per query and for each condition, so that the two conditions can be paired.

Table 3. Outcome measures, operational definitions, and levels of measurement

Measure	Operational definition	Per-query measurement	Aggregate
Harmful Density Reduction, ΔHD (SOP1, H1)	Proportion of documents labeled HD in the pre-filter pool D_{pre} ($k = 50$) minus the proportion in the post-filter pool D_{post} ($k \leq 30$)	Proportion (paired difference from -1 to 1)	Median and mean of ΔHD
Ground-Truth Retention Loss, GTRL (SOP2, H2)	Proportion of ground-truth supportive documents (GD) present in D_{pre} that are removed by Stage 2; 0.0 when D_{pre} contains none	Proportion (0 to 1)	Median and mean of GTRL
Final Context Harmful Exposure, $HD@Top-5$ (SOP3, H3)	Number of documents labeled HD among the five passed to the generator, divided by 5 , and its matched shift $\Delta HD@Top-5 = HD(D_{pre}) - HD@Top-5$	Proportion (0 , 0.2 , ..., 1); paired difference	Median and mean of $\Delta HD@Top-5$
Unsupported Claim Rate, UCR (SOP4, H4)	Number of claims labeled Unsupported, divided by the number of claims in the answer; the decision line is excluded from claim scoring	Proportion (0 to 1)	Median and mean of UCR

Answer-level Hallucination Rate (descriptive)	The answer contains at least one claim labeled Unsupported or Contradicted	Binary (0 or 1)	Proportion of queries with $H = 1$
Harmful-Documents-Induced Hallucination Rate (descriptive)	At least one hallucinated claim is supported by a document labeled HD in that answer's context	Binary (0 or 1)	Proportion of queries with $HHD = 1$
Answer Accuracy (descriptive)	The extracted decision (yes or no) equals the PubMedQA reference label; "maybe" counts as not matching	Binary (0 or 1)	Proportion of correct answers
Hit@5 (descriptive)	At least one document labeled GD among the five passed to the generator	Binary (0 or 1)	Proportion of hits (total hits / total queries)
Filter detection (descriptive)	Confusion matrix of the filter decision (removed or retained) against the label (HD or not HD) over the 50 candidates of every test query	Per candidate	TPR, FPR, precision, F1

Hallucination Scoring Protocol

The protocol adapts claim-level evaluation as used in FActScore, RAGAs, and RAGTruth (Min et al., 2023; Es et al., 2024; Niu et al., 2024), and is applied by the judge model according to the Hallucination Scoring Rubric. The protocol has four parts. First, the generated answer is decomposed into atomic claims, excluding the decision line,

hedging phrases, and statements about the provided documents themselves. Second, each claim is labeled against the reference evidence as Supported (entailed by the evidence), Unsupported (neither entailed nor contradicted), or Contradicted (conflicting with the evidence). Third, the answer is flagged as hallucinated when at least one claim is Unsupported or Contradicted. Fourth, in a separate pass, each hallucinated claim is checked against the documents labeled HD in the context of that same answer; if at least one of them supports the content of the claim, the hallucination is counted as harmful-document-induced.

The reference evidence for a query is the PubMedQA context and long answer of the query, its reference label, and any document in the answer's own context that is labeled GD. The judge model differs from the generator, is given no information about the condition, and scores the answers in randomized order. The attribution pass is separate so that the document labels do not influence the first three parts.

Reliability Checks

Three reliability checks quantify the noise of the measurement process. Judge consistency is assessed by re-scoring a random subset of answers with the same judge and computing the agreement between the two runs. Generator stability is assessed by regenerating answers for a random subset of queries and comparing the outcomes. Filter stability is assessed by comparing the filter across its three random seeds, reporting the mean and standard deviation of the TPR and FPR.

Filter Detection Performance (SOP1)

The detection performance of the filter is reported for the treatment condition on the labeled candidates of the test queries, with harmful as the positive class. The study reports the confusion matrix, the True Positive Rate, the False Positive Rate, Precision, and the F1-score, with 95% percentile bootstrap confidence intervals obtained by resampling test queries with replacement (Efron & Tibshirani, 1993), since candidates of the same query are not independent. The area under the ROC curve (AUROC) and the area under the precision–recall curve (PR-AUC) are reported as threshold-independent supplementary summaries. SOP1 is descriptive and has no hypothesis test.

The F1-score is used to select the threshold and to summarize detection performance for three reasons. First, harmful documents are a minority among the 50 candidates of a query, and on such class-imbalanced data raw accuracy is misleading because a filter that removes nothing would still score highly. Second, both types of error matter in this study: a false positive discards evidence that may be needed, and a false negative lets a harmful document reach the reranker, and F1, as the harmonic mean of precision and recall (van Rijsbergen, 1979), penalizes a filter that improves one at the expense of the other. Third, F1 is a standard single-number criterion for tuning a threshold on a validation set. F1 has limitations, since it ignores true negatives and depends on the threshold (Powers, 2011); it is therefore reported together with the False Positive Rate, the PR-AUC (Saito & Rehmsmeier, 2015), and the confusion matrix. F1 is not used as a measure of hallucination, which is measured by the Hallucination Rate.

Comparison of the Conditions (SOP2 and SOP3)

Because every test query is processed by both conditions, all comparisons are paired by query. Table 4 shows the pre-specified test for each outcome. Binary outcomes are analyzed with McNemar's test (McNemar, 1947), which is appropriate for paired categorical data. For the continuous outcome, the paired t-test is the primary test when the paired differences are consistent with normality, and a paired permutation (sign-flip) test is used otherwise. Normality is assessed with the Shapiro–Wilk test on the paired differences (Shapiro & Wilk, 1965) at a significance level of 0.05, supplemented by inspection of histograms and Q–Q plots; the decision rule is fixed before the results are seen, so that the choice of test cannot depend on the outcome. Rank-based tests such as the Wilcoxon signed-rank test and their effect sizes are not used.

Table 4. *Summary of Pre-Specified Statistical Tests*

Outcome Measure	Level of Measurement	Statistical Test	Reported Statistics
Harmful Density Reduction, Δ HD (H1, SOP1)	Paired continuous (proportions, 0.0 to 1.0)	Paired Wilcoxon Signed-Rank Test with Pratt's zero-method (directed one-tailed, Median(Δ HD) > 0)	N, W (or Z), p-value, sample mean and median of Δ HD
Ground-Truth Retention Loss, GTRL (H2, SOP2)	Continuous (proportion, 0.0 to 1.0)	One-Sample Wilcoxon Signed-Rank Test with Pratt's zero-method against $\delta = 0.05$ (directed one-tailed, Median(GTRL) < 0.05)	N, W (or Z), p-value, sample mean and median of GTRL

HD@Top-5 shift, Δ HD@Top-5 (H3, SOP3)	Paired continuous (proportions, 0.0 to 1.0)	Paired Wilcoxon Signed-Rank Test with Pratt's zero-method (directed one-tailed, Median(Δ HD@Top-5) > 0)	N, W (or Z), p-value, sample mean and median of Δ HD@Top-5
Unsupported Claim Rate, UCR (H4, SOP4)	Continuous (proportion, 0.0 to 1.0)	One-Sample Wilcoxon Signed-Rank Test with Pratt's zero-method against $\theta = 0.10$ (directed one-tailed, Median(UCR) < 0.10)	N, W (or Z), p-value, sample mean and median of UCR

All tests are two-sided at a significance level of 0.05, and a hypothesis is supported only if the observed difference is in the direction stated. H1 is the only confirmatory hypothesis and addresses the objective of the study through a single primary endpoint, so no correction for multiple comparisons is applied to it. H2, H3, and H4 are supporting hypotheses; their results are reported with exact p-values and confidence intervals, and are interpreted as evidence on the mechanism and on the cost of filtering and not as separate confirmatory claims. H2 is considered supported only if both the Harmful-Document-Induced Hallucination Rate and the Answer Accuracy differ in the stated direction. Because 92.8% of the PubMedQA labels in this subset are “yes” (Jin et al., 2019), Answer Accuracy is interpreted against the majority-class rate, and it is also reported separately for queries with the reference label “yes” and “no”.

Statistical Treatment

The statistical analysis evaluates whether the proposed pre-reranking safety filter significantly reduces hallucination and harmful document exposure without compromising factual retrieval. All hypothesis tests are two-sided and evaluated at a significance level of $\alpha = 0.05$.

Because every test query traverses the same pipeline, and the pre-filter, post-filter, and final-context measurements are all computed for the same query, the resulting metrics are statistically dependent (matched pairs). Consequently, tests designed for independent observations, such as the standard Chi-Square test of independence, are invalid for this study. The statistical treatment exclusively employs matched-pair and one-sample testing frameworks to correctly account for within-query correlation.

A. Non-Parametric Wilcoxon Signed-Rank Test

Biomedical retrieval metrics, such as passage counts, density ratios, and claim proportions, systematically violate the assumptions of normality, homoscedasticity, and continuous differentiability required by parametric tests such as the paired t-test. Specifically, document distributions exhibit severe floor effects (for example, zero harmful documents retrieved for benign queries) and bounded support in $[0, 1]$.

Consequently, all inferential evaluations are conducted exclusively with the non-parametric Wilcoxon Signed-Rank Test (Wilcoxon, 1945). Hypotheses 1 and 3 are evaluated with the Paired Wilcoxon Signed-Rank Test across matched query

observations. Hypotheses 2 and 4 are evaluated with the One-Sample Wilcoxon Signed-Rank Test against fixed clinical benchmark thresholds ($\delta = 0.05$ and $\theta = 0.10$, respectively).

B. Pratt's Zero-Method Formulation

A substantial proportion of clinical queries naturally contain zero harmful documents in both the pre-filter and post-filter pools, which yields a paired difference of exactly zero ($D_i = X_{2,i} - X_{1,i} = 0$). Standard Wilcoxon conventions, such as the Wilcoxon procedure, drop all zero-difference pairs prior to ranking. In safety-critical evaluations, discarding ties artificially deflates the effective sample size (N) and introduces selection bias by censoring the queries in which the pipeline operated with zero defects.

To preserve complete sample fidelity, this study adopts Pratt's zero-method (Pratt, 1959). Under Pratt's formulation, zero-difference pairs are retained during the absolute ranking phase and are eliminated only when computing the positive and negative signed rank sums. The procedure has three steps.

1. Compute matched differences across all N evaluated queries: $D_i = X_i - Y_i$ (or $D_i = X_i - \mu_0$ for one-sample tests)
2. Rank all N pairs based on the absolute magnitudes $|D_i|$, assigning average ranks to tied values: $R_i = \text{Rank}(|D_i|)$, for all i in $\{1, 2, \dots, N\}$ Pairs where $D_i = 0$ receive the lowest ranks, corresponding to ties at zero.
3. Compute the signed rank sums W^+ and W^- by partitioning the ranks according to the sign of the original non-zero difference: $W^+ = \sum R_i$ over all i with $D_i > 0$, $W^- = \sum R_i$ over all i with $D_i < 0$

The test statistic $W = \min(W+, W-)$ is evaluated against the exact Wilcoxon distribution for $N \leq 50$, and against the continuity-corrected asymptotic normal approximation for large samples ($N > 50$):

$$Z = (W - N(N+1)/4) / \sqrt{ ([N(N+1)(2N+1) - \sum(t^3 - t)] / 24)}$$

where t represents the count of tied ranks within each tie cluster. For paired tests, $D_i = X_i - Y_i$ is the difference between the pre-filter and post-filter measurements of query i (Hypotheses 1 and 3); for one-sample tests, μ_0 is the fixed benchmark, $\delta = 0.05$ for Hypothesis 2 and $\theta = 0.10$ for Hypothesis 4.

C. Decision Rules and Reporting Criteria

All statistical hypothesis tests are conducted at an a priori significance threshold of $\alpha = 0.05$. A null hypothesis $H_{0,k}$ is formally rejected if the calculated p-value satisfies $p < 0.05$. The four hypotheses address distinct operational scopes (Stage 2 density, Stage 2 retention, Stage 3 context exposure, and Stage 4 hallucination), each with its own metric, and each is tested once; no adjustment for multiplicity is applied. In accordance with non-parametric reporting standards, the inferential output for each hypothesis explicitly presents the following.

- The total evaluated paired sample size (N).
- The test statistic (W or Z).
- The exact two-tailed (or directed one-tailed) p-value.
- The observed sample mean and median shift values.

D. Methodological Rationale for Analytical Streamlining

Prior biomedical NLP studies frequently attempt to combine multiple competing inferential procedures, such as tests that require binarized outcomes, pseudo-median estimators, resampling-based interval estimates, and stepwise adjustments of significance levels. In this study, those procedures are excluded from the methodology for three reasons.

1. **Elimination of information loss from artificial dichotomization.** Tests for paired categorical data require continuous proportional metrics, such as harmful density or retention loss, to be artificially binarized into arbitrary pass/fail indicators. Binarizing continuous clinical densities destroys nuance and discards statistical power. The Wilcoxon Signed-Rank Test natively accommodates continuous and interval data without loss of precision.
2. **Exact non-parametric power without distributional assumptions.** Resampling-based interval estimates and pseudo-median estimators introduce stochastic resampling variance and assume symmetrical difference distributions around a pseudo-median. In contrast, the Wilcoxon test with Pratt's zero-method yields exact, distribution-free p-values that directly evaluate whether matched shifts are stochastically dominant.
3. **Guaranteed preservation of complete query cohorts.** Stepwise significance adjustments and arbitrary effect size bounds overparameterize the analytical pipeline. By anchoring the analysis strictly to four orthogonal research questions with distinct operational scopes (Stage 2 density, Stage 2 retention, Stage 3 context exposure, and Stage 4 hallucination), the single-test Wilcoxon protocol delivers transparent, defensible, and reproducible clinical conclusions.

E. Filter Classification Performance

To describe how well the Stage 2 cross-encoder filter balances blocking harmful documents against incorrectly dropping safe ones, the filter's isolated classification performance is evaluated on the test set. With Harmful Documents (HD) designated as the positive class, the following metrics are computed and reported descriptively:

- **True Positive Rate (Recall / Sensitivity):** $TPR = TP / (TP + FN)$
- **False Positive Rate:** $FPR = FP / (FP + TN)$
- **Precision:** $Precision = TP / (TP + FP)$
- **F1-Score:** $F1 = 2 \times (Precision \times TPR) / (Precision + TPR)$

Threshold-independent discrimination is additionally summarized via AUROC and PR-AUC.

F. Inter-Rater Reliability

Cohen's Kappa (κ) is used to evaluate inter-rater agreement for both candidate document annotation (LLM labeler vs. human reviewers) and hallucination scoring (judge model vs. human reviewers):

$$\kappa = (p_o - p_e) / (1 - p_e)$$

where p_o is the observed proportional agreement and p_e is the hypothetical probability of chance agreement. A benchmark criterion of $\kappa \geq 0.80$ is required to establish substantial-to-near-perfect reliability prior to frozen experimental runs (Landis & Koch, 1977).

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APPENDIX A: EXPERIMENT PAPER

This sheet is completed by the researchers for every experimental run and is kept with the run outputs. It records the procedure that was followed, the fixed parameters, and every deviation from the plan, so that the experiment paper can be properly documented, audited, and reproduced. Items in brackets are completed at the time of the run.

Table 5. Filter Detection Performance on Test Candidates (SOP 1)

Metric	Value	95% CI (bootstrap)
True Positive Rate (Recall)		
False Positive Rate		
Precision		
F1-Score		
AUROC		
PR-AUC		

Note. Harmful (HD) is the positive class. Confidence intervals from $B = 10,000$ bootstrap resamples of test queries.

Table 6. Confusion Matrix of Filter Decisions vs. Document Labels (SOP 1)

	Predicted Harmful (removed)	Predicted Not Harmful (retained)
Actual HD		
Actual GD/MD		

Note. $n = __$ labeled candidates across 500 test queries (50 candidates each).

Table 7. Filter Stability Across Random Seeds

Seed	TPR	FPR
Seed 1		
Seed 2		
Seed 3		
Mean $\pm$ SD		

Table 8a. Binary Paired Outcomes — Rates and Effect Size (H1, H2, H4)

Outcome	Control Rate	Treatment Rate	Δp (95% CI)
Hallucination Rate (H1, confirmatory)			
Harmful-Documents-Induced Hallucination Rate (H2)			
Answer Accuracy (H2)			
GD@Top-5, Hit@5 (H4)			

Note. Positive Δp indicates improvement under Treatment. CI from 95% percentile bootstrap.

Table 8b. Binary Paired Outcomes — McNemar Test Details (H1, H2, H4)

Outcome	b (Control+/Treatment-)	c (Control-/Treatment+)	Test Statistic	p-value

Hallucination Rate (H1, confirmatory)				
Harmful-Documents-Induced Hallucination Rate (H2)				
Answer Accuracy (H2)				
GD@Top-5, Hit@5 (H4)				

Note. McNemar's test with Edwards' correction reported when $b + c \geq 25$; exact binomial p-value substituted otherwise — flag which was used per row.

Table 9. Answer Accuracy by Reference Label (Supplementary to H2)

Reference Label	n	Control Accuracy	Treatment Accuracy	Δp	p-value
"Yes"					
"No"					
Overall					

Table 10a. Continuous Paired Outcomes — Descriptive Statistics (H3 and Secondary)

Outcome	Control M (SD)	Treatment M (SD)	Mean difference $\bar{d}$ (95% CI)
HD@Top-5 (H3)			
Unsupported-claim rate (secondary)			

Note. CI from 95% percentile bootstrap ($B = 10,000$).

Table 10b. Continuous Paired Outcomes — Test Details (H3 and Secondary)

Outcome	Shapiro-Wilk p	Test Used	Test Statistic	p-value	Effect Size
HD@Top-5 (H3)		Paired t-test / Permutation			
Unsupported-claim rate (secondary)		Paired t-test / Permutation			

Note. Paired t-test used if Shapiro–Wilk $p > 0.05$ (normality holds); paired permutation (sign-flip) test used otherwise — indicate which per row.

Table 11. Inter-Rater Reliability (Cohen's Kappa)

Agreement Check	Rates Compared	n pairs/answers	κ	Interpretation
Document annotation	LLM labeler vs. human reviewers	300		
Hallucination scoring	Judge model vs. human reviewers	100 per condition (200 total)		

Note. Benchmark $\kappa \geq 0.80$ required prior to frozen experimental runs (Landis & Koch, 1977).

Table 12. Reliability Checks on the Measurement Process

Check	What is Compared	Metric	Result
Judge consistency	Same judge, re-scored subset	Agreement rate	
Generator stability	Regenerated answers, same queries	Outcome agreement	
Filter stability	Across 3 seeds	Mean/SD of TPR, FPR	See Table 7

Table 13. Summary of Hypothesis Test Outcomes

Hypothesis	Direction Predicted	Result	Supported
H1 (primary): Treatment has lower Hallucination Rate	Treatment < Control		(Yes/No)
H2: Treatment has lower HDIHR and non-inferior Accuracy	Both conditions must hold		(Yes/No)
H3: Treatment has lower HD@Top-5	Treatment < Control		(Yes/No)
H4: GD@Top-5/Hit@5 differs between conditions	Two-sided		(Yes/No)

APPENDIX B: RESEARCH INSTRUMENT OUTLINES

B1. Document Annotation Guide (Codebook)

The guide is given to the labeling model and to the human reviewers. For each (query, document) pair, the rater is shown the query, the reference answer (the long answer) with its reference label, and the document; the retriever rank and the pipeline condition are not shown. Table 14 gives the label definitions, and the decision procedure follows.

Table 14. *Label definitions in the annotation guide*

Label	Definition	Worked examples
Ground-truth (GD)	The document directly and consistently supports the reference answer, that is, its conclusion and its yes or no label.	
Harmful (HD)	The document shares the topic, entities, or wording of the query but does not support the reference answer. It contradicts the reference answer, supports the opposite conclusion, or addresses a different population, intervention, or outcome, so that relying on it would produce an unsupported or wrong answer.	
Mediocre (MD)	The document is partially relevant or neutral. It provides related background or incomplete information that neither supports the reference answer nor pushes toward an unsupported or wrong answer.	

1. Read the query, the reference answer, and the reference label.
2. Read the document in full.
3. If the document directly supports the reference answer, assign GD.
4. If the document shares the topic of the query but would lead a reader to an unsupported or opposite conclusion, assign HD.
5. Otherwise, assign MD.
6. Write a one-sentence rationale that cites the passage that determined the label.

Boundary rules for cases in which a document supports only part of the reference answer, is off-topic, or falls between two labels are finalized during the pilot and added to the guide.

B2. Document Annotation Sheet

Table 15. *Columns of the document annotation sheet*

Column	Content
Pair ID and Query ID	Identifiers of the (query, document) pair and of the query
Query and reference answer	The query text, the long answer, and the reference label
Document ID and text	Identifier and full text of the candidate document (retriever rank hidden from raters)
Labeling-model label and rationale	GD, HD, or MD, with a one-sentence rationale
Reviewer ID, label, and rationale	Independent human label and rationale, recorded under an anonymous reviewer identifier

Adjudicated label	Final label after resolution of disagreements, with the date
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B3. Hallucination Scoring Rubric

Table 16. *Elements of the hallucination scoring rubric*

Element	Definition or rule
Decision line	The line beginning “Decision:” is excluded from claim scoring and is scored only by Answer Accuracy against the reference label.
Claim	A single, self-contained factual statement about the biomedical content of the explanation. Compound sentences are split, pronouns are resolved, and questions, hedging phrases, and statements about the sufficiency of the provided documents are excluded.
Supported	The claim is entailed by the reference evidence.
Unsupported	The claim is neither entailed by nor contradicted by the reference evidence.
Contradicted	The claim conflicts with the reference evidence.
Reference evidence	The PubMedQA context and long answer of the query, its reference label, and any document in the answer’s own context that is labeled GD.
Answer-level flag	The answer is hallucinated if at least one claim is Unsupported or Contradicted.
Attribution rule	For each Unsupported or Contradicted claim, the documents labeled HD in the same answer’s context are checked. If at least one of them supports the content of the claim, the hallucination is

	harmful-document-induced. This is done in a separate pass.
Blinding	Answers are coded, stripped of any condition identifier, and presented in randomized order.

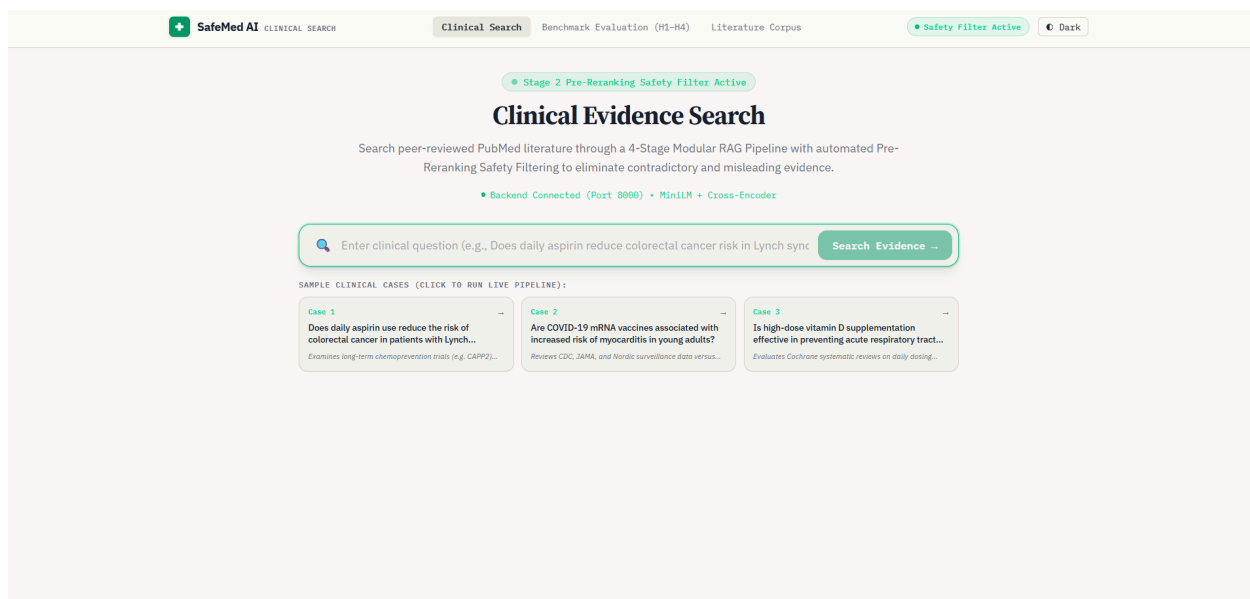
B4. Hallucination Scoring Sheet

Table 17. Columns of the hallucination scoring sheet

Column	Content
Answer code and Query ID	Random code of the answer (no condition information) and the identifier of the query
Extracted decision	yes, no, or maybe, and whether it equals the reference label
Claim ID and claim text	Each atomic claim extracted from the explanation
Claim label and evidence pointer	Supported, Unsupported, or Contradicted, with the part of the reference evidence that decided the label
Answer-level flag	1 if at least one claim is Unsupported or Contradicted, otherwise 0
Attribution	Identifiers of the HD documents that support each hallucinated claim, and the resulting harmful-document-induced flag
Rater and notes	Judge model or anonymous human reviewer identifier, date, and comments

APPENDIX C: SYSTEM/TOOL MOCKUP

The screens in this appendix are conceptual mockups of a user-facing application that could present the output of the pipeline. They illustrate the interface concept only. The content shown, including the example answer and the source labels, is illustrative and is not an output of the evaluated system; the evaluation in this study concerns the retrieval pipeline and the answers it generates, and no clinical validation is claimed.



SAMPLE CLINICAL CASES (CLICK TO RUN LIVE PIPELINE): [Reset](#) [Search](#)

Case 1

Does daily aspirin use reduce the risk of colorectal cancer in patients with Lynch...

Proven long-term chemoprevention trials (e.g. CAPP2)...

Case 2

Are COVID-19 mRNA vaccines associated with increased risk of myocarditis in young adults?

Review CDC, MMWR, and Morbidity and Mortality data sources...

Case 3

Is high-dose vitamin D supplementation effective in preventing acute respiratory tract...

Proven chemoprevention studies on daily dosing...

✓ CLINICAL CONSENSUS Verified Clinical Consensus 50 Passages Screened 5 Blocked Harmful Top-5 Safe Context

CLINICAL QUESTION:

Does daily aspirin use reduce the risk of colorectal cancer in patients with Lynch syndrome?

EVIDENCE SYNTHESIS & FINDINGS:

Based on verified biomedical literature, clinical findings indicate: Long-term follow-up of the CAPP2 randomized controlled trial demonstrates that daily aspirin (600 mg/day) taken for an average of 25 months significantly reduces the incidence of colorectal cancer in patients with Lynch syndrome by approximately 40% to 50% after 10 years. International clinical guidelines recommend regular aspirin chemoprevention for patients diagnosed with hereditary nonpolyposis colorectal cancer (Lynch syndrome), taking into account patient GI bleeding risk. Individual participant data meta-analysis confirms that daily or weekly vitamin D supplementation is safe and protects against acute respiratory tract infections, particularly among individuals who are baseline vitamin D deficient.

Top-5 Evidence Sources (5) Pre-Ranking Safety Log Blocked Atomic Claims (2)

Thesis Evaluation Breakdown

These 5 verified passages were selected by the Stage 2 Cross-Encoder Ranker after the Stage 2 Safety Filter purged all harmful/contradictory candidates:

[1] CAPP2 Long-Term Follow-up Trial ✓ VERIFIED SAFE

The Lancet - Cross-Encoder Relevance: 10.2116

"Long-term follow-up of the CAPP2 randomized controlled trial demonstrates that daily aspirin (600 mg/day) taken for an average of 25 months significantly reduces the incidence of colorectal cancer in patients with Lynch..."

[2] Aspirin Chemoprevention in Lynch Syndrome Guidelines ✓ VERIFIED SAFE

Gastroenterology - Cross-Encoder Relevance: 7.1567

"International clinical guidelines recommend regular aspirin chemoprevention for patients diagnosed with hereditary nonpolyposis colorectal cancer (Lynch syndrome), taking into account patient GI bleeding risk..."

[3] BMJ Systematic Review on Vitamin D Supplementation ✓ VERIFIED SAFE

BMJ - Cross-Encoder Relevance: -8.3490

"Individual participant data meta-analysis confirms that daily or weekly vitamin D supplementation is safe and protects against acute respiratory tract infections, particularly among individuals who are baseline vitamin D..."

[4] CDC Morbidity & Mortality Weekly Report ✓ VERIFIED SAFE

CDC MMWR - Cross-Encoder Relevance: -8.8254

"Review of safety data showing myocarditis cases of 10-15 cases per million doses in young males, benefits of vaccination against COVID-19 hospitalization and death far outweigh the rare risk of myocarditis..."

[5] Diabetes Prevention Program Study - Cohort 11 ✓ VERIFIED SAFE

N Engl J Med - Cross-Encoder Relevance: -9.8012

"Metformin therapy in prediabetic adults reduces the 5-year progression to type 2 diabetes by 29% compared to placebo. Additional multi-center observational analysis confirmed robust external validity across 1,000 participants..."

SafeMed AI

CLINICAL SEARCH

Clinical Search

Benchmark Evaluation (H1-H4)

Literature Corpus

Safety Filter Active

Dark

Thesis Benchmark Evaluation (PubMedQA)

Standardized Held-Out Test Split • Matched-Pairs Comparative Analysis

Pilot Test Split • 5 Queries Evaluated

METHODOLOGY: 4-STAGE PRE-RERANKING FILTER ARCHITECTURE

Standard RAG fails because cross-encoder rerankers elevate topically similar yet contradictory documents. SafeMed AI intercepts candidates at **Stage 2 (Pre-Reranking)** via a fine-tuned Cross-Encoder, dropping harmful literature before Stage 3 selects the final Top-5 context. All inferential statistics are evaluated using **Wilcoxon Signed-Rank Tests** with **Pratt's zero-method**.

Primary Hypothesis Tests (H1 – H4)

Evaluated at $\alpha = 0.05$ (two-tailed) on within-query matched pairs.

H1 (SOP 1)

Fail to reject H_0

Harmful Document Exposure Reduction

Paired Wilcoxon test on Control vs. SafeMed AI HD@Top-5

METRIC	CONTROL (VANILLA)	SAFEMED AI
Mean HD@Top-5 Exposure	12.0%	↓ 0.0%
Net HER Reduction		12.0%

Inferential Test

Test Statistic (W)

0.00

p-value

0.250000 (Holm: 0.5000)

Mean Shift ($\hat{\mu}$)

-0.1200

Median Shift ($\hat{c}$)

-0.2000

Wilcoxon Signed-Rank Test (Paired)

H2 (SOP 2)

Fail to reject H_0

Ground-Truth Retention Rate (Hit@5)

One-sample Wilcoxon test against non-inferiority margin

METRIC	CONTROL (VANILLA)	SAFEMED AI
GD Retention Rate (Hit@5)	40.0%	↑ 24.0%

Inferential Test

Test Statistic (W)

0.00

p-value

0.500000 (Holm: 0.5000)

Mean Shift ($\hat{\mu}$)

0.2400

Median Shift ($\hat{c}$)

0.0000

Rank-Biserial (r_{rb})

1.0000

95% Bootstrap CI

[0.0000, 0.5600]

Wilcoxon Signed-Rank Test (One-Sample)

SafeMed AI CLINICAL SEARCH

Clinical Search

Benchmark Evaluation (H1-H4)

Literature Corpus

Safety Filter Active

Dark

Thesis Benchmark Evaluation (PubMedQA)

Standardized Held-Out Test Split • Matched-Pairs Comparative Analysis

- Pilot Test Split • 5 Queries Evaluated

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Harmful Document Exposure Reduction

Paired Wilcoxon test on Control vs. SafeMed AI HD@Top-5

Metric	CONTROL (VANILLA)	SAFEMED AI
Mean HD@Top-5 Exposure	12.0%	0.0% ↓
Net HER Reduction		12.0%

Inferential Test

Wilcoxon Signed-Rank Test (Paired)

Test Statistic (W)	-0.00
p-value	0.250000 (Holm: 0.5000)
Mean Shift ($\hat{\delta}$)	-0.1200
Median Shift ($\hat{m}$)	-0.2000

Fail to reject H_0

H2 (SOP 2)

Ground-Truth Retention Rate (Hit@5)

One-sample Wilcoxon test against non-inferiority margin

Metric	CONTROL (VANILLA)	SAFEMED AI
GD Retention Rate (Hit@5)	40.0%	24.0% ↓

Inferential Test

Wilcoxon Signed-Rank Test (One-Sample)

Test Statistic (W)	0.00
p-value	0.500000 (Holm: 0.5000)
Mean Shift ($\hat{\delta}$)	0.2400
Median Shift ($\hat{m}$)	0.0000
Rank-Biserial (r_{rb})	1.0000
95% Bootstrap CI	[0.0000, 0.5600]

Fail to reject H_0